DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

SCHOOL OF SCIENCE

DEPARTMENT OF MATHEMATICS AND PHYSICAL SCIENCES

SPH 2171 PHYSICS II SPH 2174 PHYSICS FOR ENGINEERS II

PART I: ELECTROSTATICS

LECTURE I: CHARGE AND COULOMB'S LAW

I.1. LECTURE OUTLINE

Electrostatics is the study of interaction of electric charges at rest, stationary charges. Electric charge is one of the fundamental proprieties of matter. That propriety manifests itself in that two particles that carry charge interact. We can distinguish two types of charges: positive and negative charges. The charge is quantized and conserved. In terms of materials we can distinguish conductors, materials that conduct electricity and insulators, materials that in normal condition can not conduct electricity.

The interaction of charges is governed by the famous Coulomb's law. The force of interaction obeys the law of superposition therefore Coulomb's law can be used to describe interaction of extended charged bodies

I.1.1 SPECIFIC OBJECTIVES

At the end of this lecture, you should able to:

1. Describe electric charge and its main propieties
2. Describe interaction between two point charges
3. State and apply Coulomb's law for point charges and certain disbution of charges

I.2. ELECTRIC CHARGE

I.2.1. INTRODUCTION: ELECTROMAGNETISM

The term ***"Electromagnetism"*** is a combination of electrical and magnetic phenomena. The word comes from two words: ***"electron"*** and ***"magnesia***"

Electron "$\varepsilon\lambda\varepsilon\kappa\tau\rho o\nu$" means "***amber***": if you rub a piece of amber with a piece of cloth, it will attract bits of straw. This is ***amber effect*** or ***electrostatic effect***.

Magnesia is the home region of ***magnetites***, somewhere in Asia Minor, in Turkey of the modern world. Magnetites are stones that attract iron.

Electromagnetism developed thanks to efforts of a lot of people and generations.
Electromagnetic force is unlimited in range, like gravity. It acts on particles with electric charge. The electrical and magnetic forces were unified into a single theoretical framework in the 19^{th} Century. Electromagnetism is the fundamental interaction that binds electrons to the nuclei to form atoms and binds atoms together in molecules and solids. It is responsible for the proprieties of solids, liquids and gases and forms the basis of the sciences of chemistry and biology. It is the fundamental interaction behind all microscopic contact forces such as the frictional forces between surfaces in contact and forces exerted by springs, muscles... elastic forces. The electromagnetic force is much stronger than gravitational force. For example, the electrical repulsion of two electrons at rest is about 10^{43} times as strong as their gravitational attraction between them.

I.2.2. TWO OPPOSITE TYPES OF CHARGES

From Greek ancient times, it was known that a rubbed piece of amber attracts bits of straw. It can be shown that there are two kinds of charge by rubbing a glass rod with silk and hanging it from a long silk thread. If a second glass is rubbed with silk and held rubbed end of the first rod, the rods repel each other as shown in Fig.I.1.

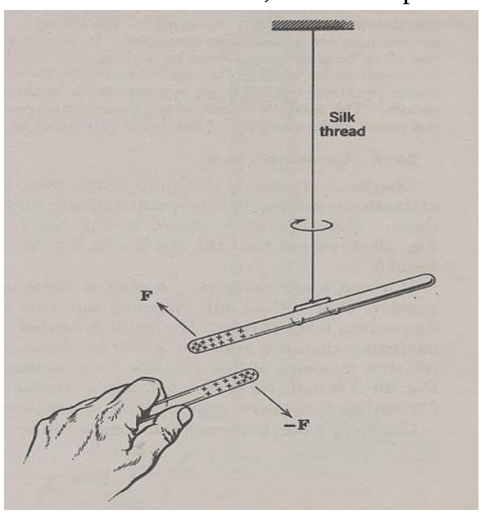


Fig.I.1. Two positively charged glass rods repel each other

On the other hand, a hard –rubber rod rubbed with fur will attract the glass rod. Two hard – rubber rods rubbed with fur will repel each other. We say that rubbing a rod gives it an ***electric charge*** and the charges exert forces on each other. Charges on the glass and on the hard-rubber are different in nature.

Like charges repel each other and unlike charges attract

B. Franklin* (1706-1790) named the charges appearing on the glass ***positive***, and charges that appears on rubber, ***negative***. Notice that he could have named them the other way around

Benjamin Franklin** (January 17, 1706 – April 17, 1790)** was one of the ***Founding Fathers of the United States. A noted ***polymath***, Franklin was a leading ***author, printer, political theorist, politician, postmaster, scientist, musician, inventor, satirist, civic activist, statesman, and diplomat.*** As a scientist, he was a major figure in the ***American Enlightenment*** and the history of physics for his discoveries and theories regarding electricity. He invented the ***lightning rod***, ***bifocals,*** the ***Franklin stove,*** a ***carriage odometer,*** and the ***glass 'armonica'***. He facilitated many civic organizations, including a fire department and a university, ***University of Pennsylvania.***

Any substance rubbed with any other under suitable conditions will become charged; by comparing the unknown charge with a charged glass rod, it can be labelled as either positive or negative.
The modern view is that, in its normal or neutral state, ***bulk matter contains equal amounts of positive and negative electricity***. If two bodies like glass and silk are rubbed together, a small amount of charge is transferred from one to the other, upsetting the electric neutrality. In this case the glass becomes positive and silk negative.

I.2.3. CONDUCTORS AND INSULATORS

A metal rod held in the hand and rubbed with silk will not seem to develop a charge. It is possible to charge such a rod if it furnished with a glass or hard-rubber handle and if the metal is not touched with the hands while rubbing it.
The explanation is that metals, the human body and the earth are ***conductors of electricity*** and that glass, hard-rubber, plastics, are ***insulators*** or ***dielectrics***.
In conductors, electric charges are free to move through the material; in insulators they are not.
In metals only negative charge is free to move: the actual charge carriers are ***free electrons*** while positive charge is as immobile as it is in glass or in any other dielectric.
In some conductors, such as ***electrolytes***, both positive and negative charge can move.
A class of materials called "***semiconductors***" is intermediate between conductors and insulators in their ability to conduct electricity. Among the elements, ***silicon*** and ***germanium*** are well known examples. Their electrical conductivity is greatly influenced when impurities are brought into their structure.

I.2.4. CHARGE IS QUANTIZED

In Franklin's days electric charge was thought of as a continuous fluid. The ***atomic theory of matter***, however, has shown that fluids themselves, such as water or air, are not continuous but are made of atoms. Experiment shows that the electric fluid is not continuous either but that it is made up of integral multiples of a certain minimum electric charge. This fundamental or ***elemental charge*** has a magnitude:

$$e = 1.602 X 10^{-19} C \tag{I.1}$$

The ***Coulomb*** (unit symbol: **C**) is the ***International System of Units (SI) unit of electric charge***. It is the charge (symbol: Q or q) transported by a constant current of one Ampere in one second:

$$1\ \mathrm{C} = 1\ \mathrm{A} \cdot 1\ \mathrm{s}$$

It is equivalent to the charge of approximately 6.242×10^{18} (1.036×10^{-5} mol) protons and −1 C is equivalent to the charge of approximately 6.242×10^{18} electrons.

Any physically existing charge q, no matter its origin, can be written as:

$$q = \pm Ne \tag{I.2}$$,

where N is an integer.

When a physical property such as charge exists in ***discrete "packets***", the property is said to be "***quantized"***
The ***quantum of charge*** e is so small that in an ordinary ***240 Volt, 100 W*** light bulb, about $2.6X10^{18}$ elementary charges e enter and leave the filament per second.

I.2.5. CHARGE AND MATTER

Actually charge is “carried” or is an ***intrinsic property*** of ***elementary particles*** that make up matter. So that when we say “charge moves”, actually the particles carrying the charge move, but what interests us is the property ”charge” of those particles. In that capacity such particles are called ”***carriers of electric charge***”.

Matter can be regarded as composed of three kinds of elementary particles: the ***proton***, the ***neutron*** and the ***electron*** shown in Table I.1.
Note the masses of the neutron and the proton are approximately equal and that the electron is lighter by a factor of about ***1840***.
Atoms are made up of a dense, positively charged ***nucleus, surrounded*** by a ***cloud of electrons***. The nucleus varies in radius from about $1X10^{-15}m$ for hydrogen and $7X10^{-15}m$ for the heaviest atoms. The outer diameter of the electron cloud: the diameter of the atom lies in the range of $3X10^{-10}m$, about 10^5 times larger than the nuclear diameter. These proportions show that actually matter is full of emptiness. The different properties of particles that make up the atom is given in Table I.1

Table I.1. Elementary particles and their electric charge

Particle	Symbol	Charge	Mass
Proton	*p*	*+e*	***1.67252 X 10^{-27} kg***
Neutron	*n*	*0*	***1.67482X 10^{-27} kg***
Electron	*e*	*-e*	***9.1091 X 10^{-31}kg***

The nucleus contains protons and neutrons, they are called ”***nucleons***”. The number of protons in the nucleus is exactly equal to the number of electrons in the electronic cloud for a normal neutral atom. This means that a normal atom and matter as a whole are ***electrically neutral***. An atom that loses an electron becomes charged positively because of the unbalanced charge of one proton; it is called a ***positive ion***. Conversely an atom which gains an additional electron becomes a ***negative ion***.

I.2.6. CHARGE IS CONSERVED

When a glass rod is rubbed with silk, a positive charge appears on the rod. Measurements show that a negative charge of equal magnitude appears on the silk. This suggests that rubbing does not create charge but merely transfers it from one object to another, disturbing slightly the electrical neutrality of each. This hypothesis is called “***principle of conservation of electrical charge***”. It has stood up under close scrutiny both for large-scale events and at the atomic and nuclear levels: no exception have been found so far. The law of conservation of electrical is one of the most important laws of matter: charge is conserved even when the particles carrying charge are not conserved!!!

I.3. COULOMB’S LAW

The law that governs the force of interaction of point charges was established experimentally in ***1785*** by the French physicist ***Charles Augustin de Coulomb****

Charles-Augustin de Coulomb* (14 June 1736 – 23 August 1806)** was a ***French physicist. He is best known for developing ***Coulomb's law***, the definition of the electrostatic force of attraction and repulsion. The ***SI unit*** of ***electric charge, the Coulomb***, was named after him.

A ***point charge***, just like a point particle, is defined as a charged body whose dimensions may be disregarded in comparison with the distances from the body to other bodies carrying an electric charge. Using a torsion balance similar to that employed by ***H. Cavendish*** to determine the gravitational constant, Coulomb measured the force of interaction of two charged spheres depending on the magnitude of the charges on them and on the distance between them.

As a result of his experiments, Coulomb arrived at the conclusion that:
"***The force of interaction between two stationary point charges is proportional to the magnitude of each of them and inversely proportional to the square of the distance between them"***

Coulomb's law can be expressed by:

$$\overline{F}_{12} = -k\frac{q_1 q_2}{r^2}\bar{e}_{12} \tag{I.3}$$

Here

- k, a ***proportionality constant*** assumed to be positive
- q_1, q_2 the ***magnitudes*** of the interacting charges
- r the ***distance*** between them
- $\bar{e}_{12}$ a ***unit vector*** directed from q_1 to q_2
- $\overline{F}_{12}$ ***force*** acting on q_1

Expression (I.3) is called "***Coulomb's law***". It is a mathematical expression that reflects all aspects of interaction of two static point charges

Note the striking similarity with the Newton's law of gravity:

$$\vec{F}_{12} = G\frac{m_1 m_2}{r^2}\bar{e}_{12}$$

Because interaction forces are proportional to the inverse of square of distance between objects, these two laws are called "***inverse-square***" laws

Experiments show that the force of interaction between two given charges does not change if other charges are placed near them. Assume that we have a charge q_a and N other charges $q_1,.q_2 \dots q_N$. The resultant force $\overline{F}$ with which all the N charges q_i act on q_a is:

$$\overline{F} = \sum_{i=1}^{N}\overline{F}_{ai} \tag{I.4}$$

where $\vec{F}_{ai}$ is the force with which charge q_i acts on q_a in the absence of the other $N-1$ charges.

This is the ***principle of superposition of electrostatic forces*** or ***principle of independence of electrostatic forces***

Coulomb's law holds for distances from $10^{-15}m$ to at least several km. Nothing suggests that it stops being correct out of these limits.

The ***SI unit*** of charge is the ***Coulomb,*** C

A ***Coulomb*** is defined as "***the amount of charge that flows through a given cross section of a wire in*** $1s$ ***if there is a steady current of*** $1A$ ***in the wire"***:

$$q = It \tag{I.5}$$

$$\rightarrow \quad 1C = 1A.1s \tag{I.6}$$

Thus, in expression (I.3), for SI unit system $[q] = Coulomb$, $[r] = m$, $[F] = N$, and constant k

$$k = \frac{1}{4\pi\varepsilon_0} \tag{I.7}$$

This constant has no physical meaning!!! It was introduced just to make the Coulomb's law consistent with SI unit system!!!

Numerically

$$\frac{1}{4\pi\varepsilon_0} = 9X10^9 \frac{N.m^2}{C^2} \tag{I.8}$$

where ε_0 is called "***permittivity of free space***" or ***"electrostatic constant"***

$$\varepsilon_0 = \frac{1}{36\pi X10^9} \frac{C^2}{N.m^2} = 8.85X10^{-12} \frac{F}{m} \tag{I.9}$$

Practical activity I.1

Example I.1

Find the net force on charge q_1 **due to the three others on Fig. I.2. Take** $q_1 = -5\mu C$, $q_2 = -8\mu C$, $q_3 = -15\mu C$ **and** $q_4 = -16\mu C$

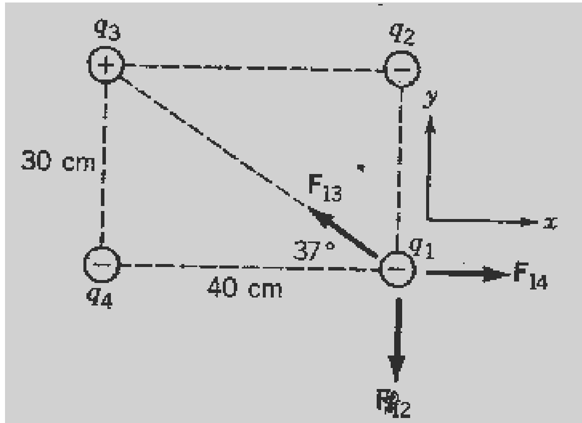


Fig.I.2. Example I.1

Solution

(i) **Numerical data:** $q_1 = -5\mu C = -5X10^{-6}C$, $q_2 = -8X10^{-6}C$, $q_3 = -15X10^{-6}C$
$q_4 = -16X10^{-6}C$, $F_1 = ?$

(ii) **Solution:** **Magnitude of forces:** The magnitude of the force on q_1 due to q_2 is**:**

$$F_{12} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1q_2|}{r^2}, \quad \text{where} \frac{1}{4\pi\varepsilon_0} = 9X10^9 \frac{N.m^2}{C^2}.$$

We can calculate and get: $F_{12} = 4N$. Similarly, we find $F_{14} = 4.5N$. Here we notice that $r_{13}{}^2 = r_{12}{}^2 + r_{14}{}^2$ and we find $F_{13} = 2.7N$

The components of the net force are:

$$F_{1X} = 0 - F_{13}\cos 37^0 + F_{14} = 2.3N$$

$$F_{1Y} = -F_{12} + F_{13}\sin 37^0 + 0 = -2.4N$$

The net force on q_1 is $\overline{F}_1 = 2.3\bar{i} - 2.4\bar{j}, N$

Answer: $\overline{F}_1 = 2.3\bar{i} - 2.4\bar{j}, N$

Summary

1. ***Electrostatics*** is the study of interaction of electric charges at rest.
2. ***Electrical charge*** is a property of matter that causes it to produce and experience electric and magnetic effects.
3. There are two types of charges: ***positive and negative*** charges and in nature the amount of both types of charge is the same. Thus the ***universe*** as a whole is ***electrically neutral***.

4. ***4.*** Electric charge obeys the law of ***conservation of charge***: the net charge on an isolated system as for the entire universe, is constant.
5. ***5.*** Charge is ***quantized:*** it appears only in discrete packets composed by an ***integer number*** of a minimum charge called ***quantum of charge*** or ***elementary charge*** such that any charge q is represented as:

$$q = \pm Ne,$$

where N is an integer and $e = 1.6X10^{-19}C$ is the elementary charge
6. ***Coulomb*** is the SI unit for charge
7. A ***conductor*** is a material that allows the flow of charge. In a metal the charges that move are free electrons. In ionized gases and electrlytes electrons and ions of both signs are mobile.
8. In an ***insulator*** all charges are atttached to specific sites and charge cannot flow through an insulator.
9. A pure ***semiconductor*** at $0K$ behaves like an insulator. Its ability to conduct electricity can be controlled by varying temperature, illumination or addition of impurities
10. The magnitude of ***electrostatic force*** between two stationaly point charges q and Q separated by a distance r is given by ***Coulomb's law***:

$$\overline{F}_{12} = -k\frac{q_1q_2}{r^2}\bar{e}_{12}$$

11. This is a ***central force*** as it acts along the line joining the two charges
12. The electrostatic force obeys the ***principle of linear superposition*** that means that the force between two particles is not affected by the presence of other charges and the net force on a charge acted upon by several charges is the vector sum of forces due to each charge taken separated.

Self-test I

1. Protons in a nucleus are separated by very small distances $\left(\approx 10^{-15}m\right)$. Why doesn't the nucleus fly apart because of the strong Coulomb repulsion between the protons?
2. Since the electrostatic force is so much stronger than the gravitational force, why do we not have more and frequent encounters with it?
3. A point charge q is placed at the midpoint between two equal point charges of magnitude Q. Is q in equilibrium? If so, is it stable or unstable equilibrium? Consider q to have (i) the same sign and (ii) the opposite sign to Q
4. An uncharged sphere is brought close to a point charge? Does either body experience a force? Explain
5. In what does the conduction of heat (i) resemble (ii) differ from electrical conduction?
6. A positively charged ball is bought close to a neutral isolated conductor. The conductor is then grounded while the ball is kept close. Is the conductor charged positively or negatively, or is it neutral, if (i) the ball is first taken away and then the ground connection is removed and (ii) the ground connection is first removed and then the ball taken away?
7. Does Coulomb's law hold for all charged bodies
8. A person standing on an electrically insulated platform touches a charged, electrically isolated conductor. Does this discharge the conductor completely?

Practical activity I.2

Problems I

1. Four point charges are located at the corners of a rectangle as shown in Fig.I.3. Take $Q = 4nC$. What is the resultant force on
   (i) $-2Q$
   (ii) $-3Q$

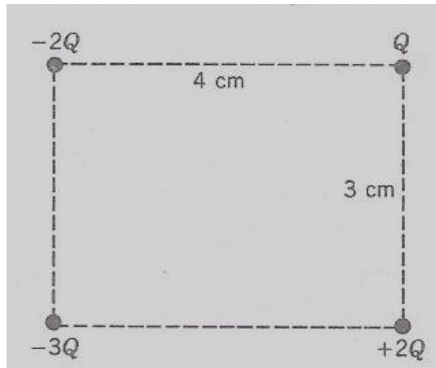


Fig.I.3. Problem I.1

2. The two balls shown in Fig.I.4 have identical masses of $0.20g$ each. When suspended from $50-cm$-long strings they make an angle of 37^0. If the charges on each are the same, how large is each charge?

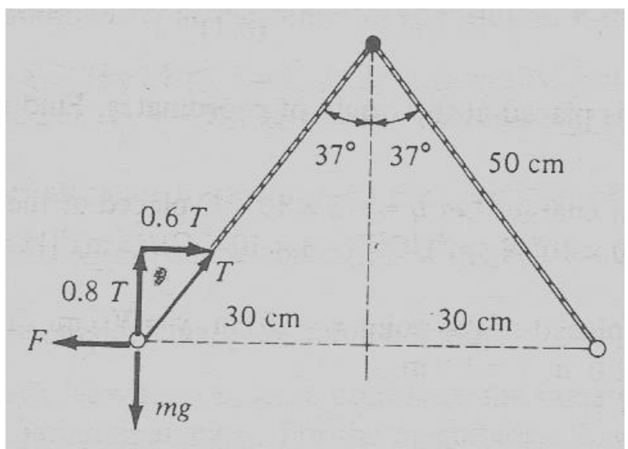


Fig.I.4. Problem I.2

3. A point charge $q_1 = 2\mu C$ is located at $(2m,1m)$ and a charge $q_2 = -5\mu C$ is at $-(2m,4m)$. Find the force exerted on q_2 by q_1
4. A charge Q is to be divided into two parts, q and $(Q-q)$ such that for a given separation the force between them is a maximum. What is the value of q?
5. Two small identical metal spheres are $3cm$ apart and attract each other with a force of $150N$. They are momentarily connected by a wire
   (i) Find the original charges if they now repel each other with a force of $10N$
   (ii) Find the original charges if the repulsive force is $150N$

LECTURE II. ELECTRIC FIELD AND ELECTRIC POTENTIAL

II.1. LECTURE OUTLINE

The interaction of charges is due to the existence of an electric field created in space around an electric charge. The strength or intensity of that electrostatic field at a point of space is measured by the force acting on a unit positive charge placed at the point. It is a vector

Electric field lines are a visual representation of the structure of the field for a given distribution of charges in space

Like the electrostatic force, the electric field obeys the principle of linear superposition.

The electrical potential is characterised by the work done by the field to move a unit positive charge. It is a scalar.

Thus the electric field and electrical potential are related. The equipotential surfaces are used to visually represent the distribution of the potential in a region of space and electric lines for a given field are normal to the equipotential surfaces.

At equilibrium, a conductor in an electric field is always an equipotential surface and electrical lines are always normal to the surface of the conductor

II.2. 1. SPECIFIC OBJECTIVES

At the end of this lecture, you should able to:

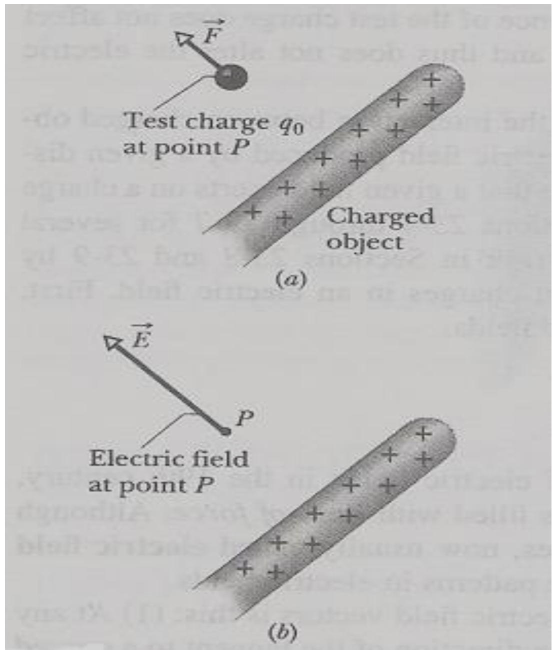
1. Define electric filed strength at point of space
2. Represent the structure of an electric field using electric field lines
3. Compute the electric field due to discrete and contiunous distribution of charge
4. Define electric filed potential at a point of space
5. Represent the structure of an electric field using field lines and equipotential surfaces
6. Compute the electric potential due to discrete and contiunous distribution of charge
7. Describe the relationship between the electric field strength and the electric potential

II.2. ELECTRIC FIELD STRENGTH

Charges at rest interact through an ***electric field***. A charge alters the properties of the space surrounding it: it sets up an electric field in it. This field manifests itself in that an electric charge placed at any point of space, experiences the action of a force. To detect and study an electric field, we use a ***test charge***.

The force acting on the test charge will characterize the field at a given point as shown in Fig.II.1.

Let us study the field set up by a stationary point charge Q with the aid of a ***point test charge*** q_0. We place the test charge at a point whose position relative to the charge Q is determined by the position vector $\bar{r}$. The test charge is usually supposed to be ***positive*** and ***enough small*** not alter the electric field it supposed to be measuring, like any good measuring instrument!!!!



(a) A positive test charge q_0 placed at point P near a charged object. An electrostatic force $\vec{F}$ acts on the test charge. (b) The electric field $\vec{E}$ at point P produced by the charged object.

Fig.II.1. A test charge is used to detect the presence of electric field

We will notice that the test charge experiences a force, F :

If we take different test charges $q_0^{/}, q_0^{//}, q_0^{///} \ldots$ then the respective forces $\vec{F}^{/}, \vec{F}^{//}, \vec{F}^{///}$ which they experience at the given point will be different.

However we will notice that the ratio $\frac{F}{q_0}$ for all the test charges is the same and is a propriety of the given point and it is called the ***electric field strength*** or ***electric field intensity*** or simply the ***electric field*** or more accurately ***electrostatic field*** at that point

$$\overline{E} = \frac{\overline{F}}{q_0} \tag{II.1}$$

It is a ***vector quantity*** directed from the point along the action of the force.

Therefore the electric field strength "***is numerically equal to the force acting a unit positive point charge at the given point of the field"***. The direction of the vector $\overline{E}$ coincides with that of the force acting on a positive charge.

Notice that electric field characterises not the charge that creates it is property of a given point of space

According to (II.1), the ***SI unit*** of the ***electric field strength*** or simply the electric field is:

$$[E]=\frac{[F]}{[Q]}=\frac{Newton}{Coulomb}\equiv\frac{1N}{C} \qquad (II.2)$$

The electric field strength set up by a point charge at a distance r will be:

$$\overline{E}=\left(\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\overline{e}_r\right) \qquad (II.3)$$

The force exerted on a test charge is:

$$\overline{F}=q_0\overline{E} \qquad (II.4)$$

If the charge q_0 is positive, the direction of the force coincides with that of the vector $\overline{E}$

We have seen that the force with which a system of charges acts on a charge not belonging to the system, equals the vector sum of the forces which each of the charges of the system exerts separately on the given charge.

Hence it follows that
"***The field strength of a system of charges equals the vector sum of the field strength that would be produced by each of the charges of the system separately***"

$$\overline{E}=\sum\overline{E}_i \qquad (II.5)$$

This statement is called the "***principle of electric field superposition***", which is a direct consequence of principle of superposition of electrostatic forces

The superposition principle allows us to calculate the field strength of any system of charges even for extended charges.

If the charge distribution is a continuous one, the field is sets up at any point P can be computed by dividing the charge into infinitesimal elements dq . The infinitesimal field $d\overline{E}$ due to each element at the point in question is then calculated treating the element as point charge. The magnitude of $d\overline{E}$ is given by:

$$d\overline{E}=\left(\frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}\right) \qquad (II.6),$$

where $\vec{r}$ is the distance from the charge element dq to the point P. The resultant field at P is then found by adding that is by integrating the field contributions due to all the charge elements:

$$\overline{E}=\int d\overline{E} \qquad (II.7)$$

Practical activity II.1

Example II.1

A thin insulating rod of length L carries a uniformly distributed charge Q . Find the field strength at a point along its axis at a distance a from the one end.

Solution

We must find the contribution to the field from an infinitesimal length element dx as shown in Fig.II.2.

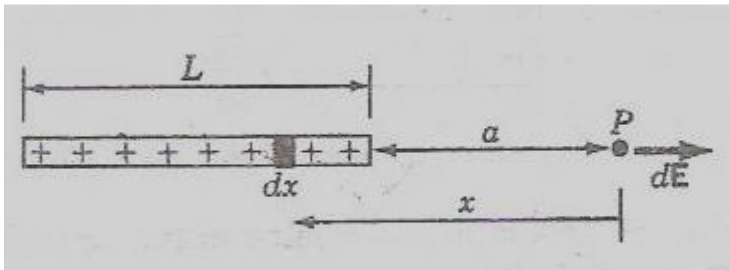


Fig.II.2 Example II.1

The charge on the element is $dq = \lambda dx$, where $\lambda = \frac{Q}{L}$ is the ***linear charge density*** or ***charge per unit length***.

The magnitude of its contribution to the field at P is: $d\overline{E} = \frac{1}{4\pi\varepsilon_0}\frac{(\lambda dx)}{x^2}$

and its direction is to the right(along x) if λ is positive.

The total field strength is the integral of $d\overline{E}$ over all elements from one of the rod to the other:

$$E = \frac{1}{4\pi\varepsilon_0}\lambda\int_a^{a+L}\frac{dx}{x^2} = \frac{1}{4\pi\varepsilon_0}\lambda\left[-\frac{1}{x}\right]_a^{a+L} = \frac{1}{4\pi\varepsilon_0}\lambda\left(\frac{1}{a} - \frac{1}{a+L}\right) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a(a+L)}$$

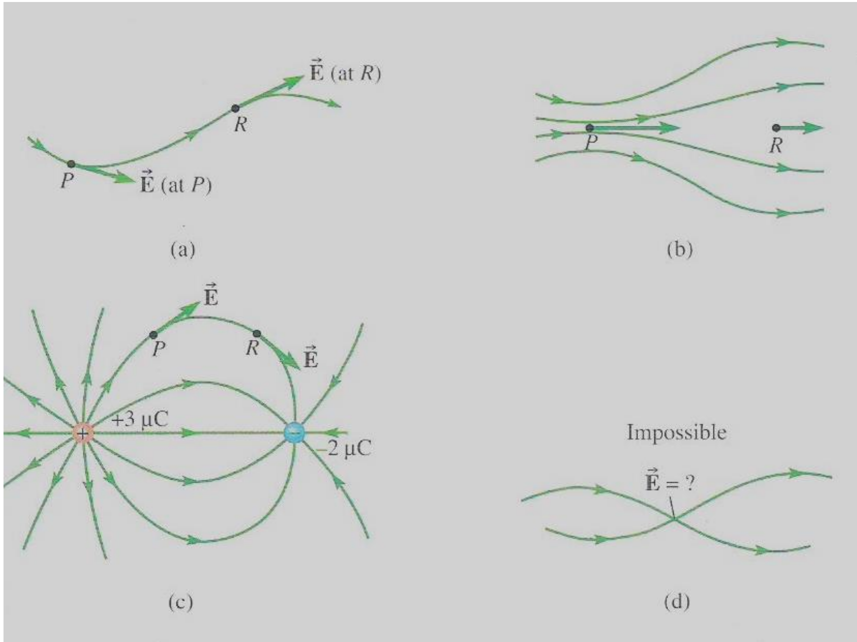
Answer: $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a(a+L)}$

This expression shows that for long distance where $a >> L$, $E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2}$

II.3. ELECTRIC FIELD LINES

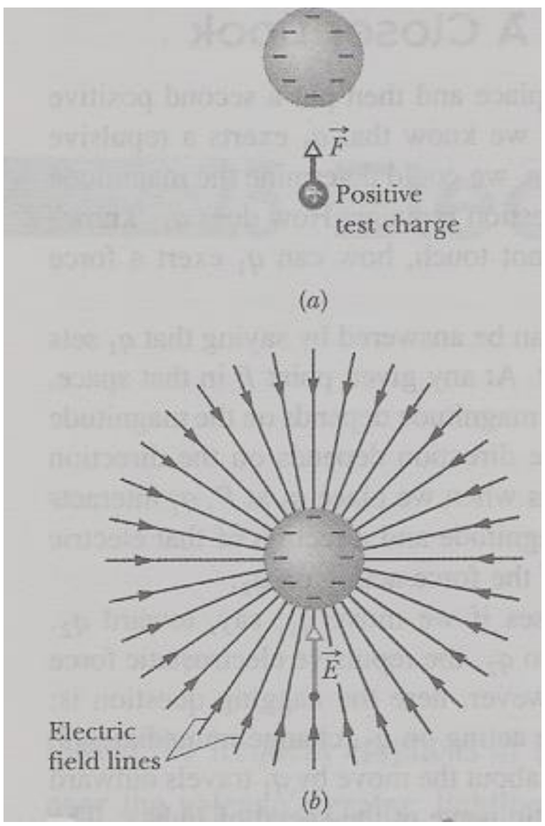
An electric field can be described by indicating the magnitude and the direction of the vector E for each point of space. The combination of these vectors forms the field of the electric field strength. It was found easier to describe at least qualitatively, an electric field with the aid of strength lines called ***electric field lines*** or ***E-lines*** or ***field lines***. This idea was introduced by ***Faraday*** himself who called them ***" lines of electric force***"!!

These lines are drawn so that a ***tangent*** to each of them at every point coincides with the ***direction*** of the vector E. The ***density of the lines*** is selected so that their number passing through a unit area at right angles to the lines equals the ***numerical value*** of the vector E. Hence the pattern of the field lines permits us to assess the direction and magnitude of vector E at various points of space. Fig.II.3(a) illustrates the concept of electric field lines and Fig.II.3(b) shows a description of field lines for some configurations of charges.

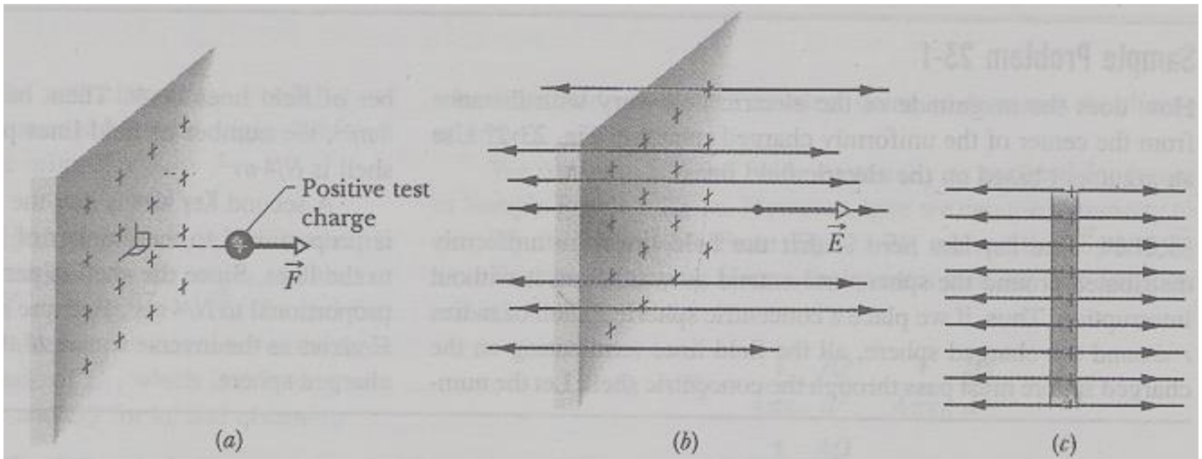


Field line rules illustrated. (a) The electric field direction at points P and R. (b) The magnitude of the electric field at point P is larger than the magnitude at R. (c) If 12 lines are drawn starting on a point charge $+3\ \mu C$, then 8 lines must be drawn ending on a $-2\ \mu C$ point charge. (d) If field lines were to cross, the direction of $\vec{E}$ at the intersection would be undetermined.

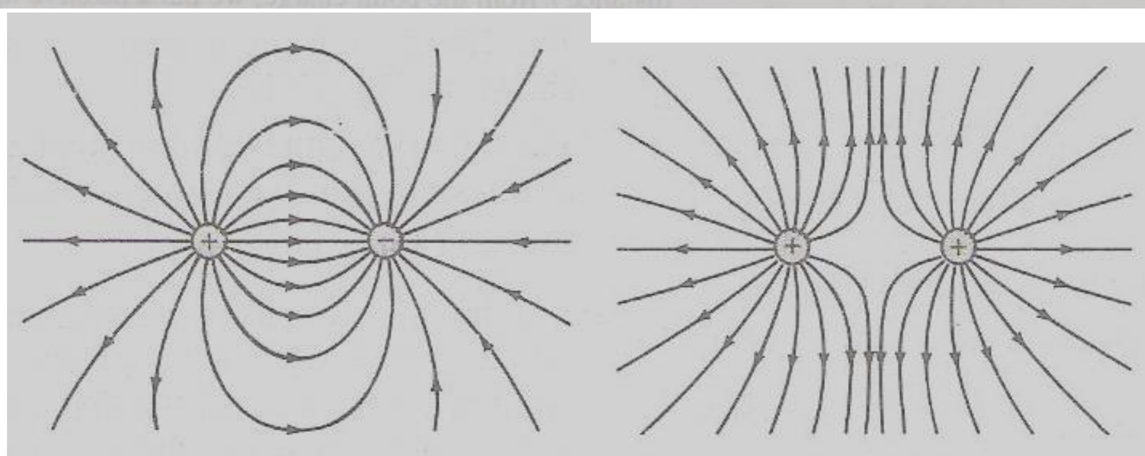
Fig.II.3 (a) Illustration of electric lines rules



(a) The electrostatic force $\vec{F}$ acting on a positive test charge near a sphere of uniform negative charge. (b) The electric field vector $\vec{E}$ at the location of the test charge, and the electric field lines in the space near the sphere. The field lines extend *toward* the negatively charged sphere. (They originate on distant positive charges.)



(*a*) The electrostatic force $\vec{F}$ on a positive test charge near a very large, nonconducting sheet with uniformly distributed positive charge on one side. (*b*) The electric field vector $\vec{E}$ at the location of the test charge, and the electric field lines in the space near the sheet. The field lines extend *away from* the positively charged sheet. (*c*) Side view of (*b*).



(c) **(d)**

Fig.II.3(b) Field lines for some configurations of charges
(c) Two equal opposite charges (d) Two equal positive charges

II.3. A POINT CHARGE IN AN ELECTRIC FIELD

An electric field exerts a force on charged particle given by (II.3): $\overline{F} = q\overline{E}$

According to Newton’s second law of motion, this force will produce acceleration: $\overline{a} = \dfrac{\overline{F}}{m}$

where m is the mass of the particle.

II.4. ELECTRIC POTENTIAL

II.4.1. ELECTRIC POTENTIAL ENERGY

Newton's law for gravitational force and Coulomb's law for the electrostatic force are mathematically identical. They must display the same features.

In particular, electrostatic force is a ***conservative force***: when an electrostatic force acts between two or more charged particles, we can assign an ***electric potential energy*** U to the system.

If the system changes its configuration from an initial state i to a different state f , the electrostatic force does work W on the particle:

$$\Delta U = U_f - U_i = -W \qquad \text{(II.8)}$$

ΔU is the resulting change in the potential energy of the system

The work done by the electrostatic force is ***path independent***:

- We take the reference configuration of a system of charged particles to be that in which the particles are all infinitely separated from each other
- We take also the corresponding reference potential energy to be zero

Let the initial potential energy U_i to be 0 and W_∞ to represent the work done by the electrostatic forces between the particles during the move in from infinity:

$$U = -W_\infty \qquad \text{(II.9)}$$

II.4.2. ELECTRIC POTENTIAL

The ***electric energy per unit charge*** is a characteristic only of the electric field investigated and does not depend on the charge q . It is called "***electric potential***" at the point:

$$\varphi = \frac{U}{q} \qquad \text{(II.10)}$$

The electric potential is a scalar, not a vector. The ***electric potential difference*** $\Delta\varphi$ between any two points i and f in an electric field is equal to the difference in potential energy per unit charge between the two points:

$$\Delta\varphi = \varphi_f - \varphi_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q} \qquad \text{(II.11)}$$

Using (II.8), equation (II.10) can be written as:

$$\Delta\varphi = \varphi_f - \varphi_i = -\frac{W}{q} \qquad \text{(II.12)}$$

If we set $U_i = 0$ at infinity as reference potential energy, we get the expression for electric potential:

$$\varphi = \frac{W_\infty}{q} \tag{II.13}$$

Thus we can define the ***electric potential*** at a point as "***the work done by electric field to move a unit positive charge from the given point to infinity"*** or equivalently ***"the work done against the electric field by external forces to move positive unit charge from infinity to the point"***

The ***SI unit***:

$$[\varphi] = \frac{[W]}{[C]} = \frac{1Joule}{1Coulomb} = \frac{1J}{1C} \equiv 1Volt = 1V \tag{II.14}$$

The unit of electric potential was named in the honour of ***Alessandro Volta****

****Alessandro Giuseppe Antonio Anastasio Volta*** **(18 February 1745 – 5 March 1827)** was an ***Italian physicist*** known for the invention of the ***electrochemical cell*** or simply the ***battery*** in the 1800s.

The ***SI unit for the electric field intensity***:

$$[E] = \frac{[F]}{[q]} = \frac{1N}{1C}\text{, but}\quad 1V = \frac{1J}{1C} \rightarrow 1C = \frac{1J}{1V} \rightarrow \frac{1N}{1C} = \frac{1N.V}{1J}\text{; but}\ \ 1J = 1N.m \rightarrow \frac{1N.V}{1N.m} = \frac{1V}{m}$$

$$\rightarrow \qquad [E] = \frac{[F]}{[q]} = \frac{1N}{1C} = \frac{1V}{1m} \tag{II.15}$$

At atomic level, energy is measured in $electronvolt \equiv 1eV$

$$1eV = 1e.XVolt = 1.6X10^{-19}CX1V = 1.6X10^{-19}J \tag{II.16}$$

Thus one ***electron-volt*** can be defined as "***the work done by or against the electric field when an electron traverses a potential difference of one volt"***

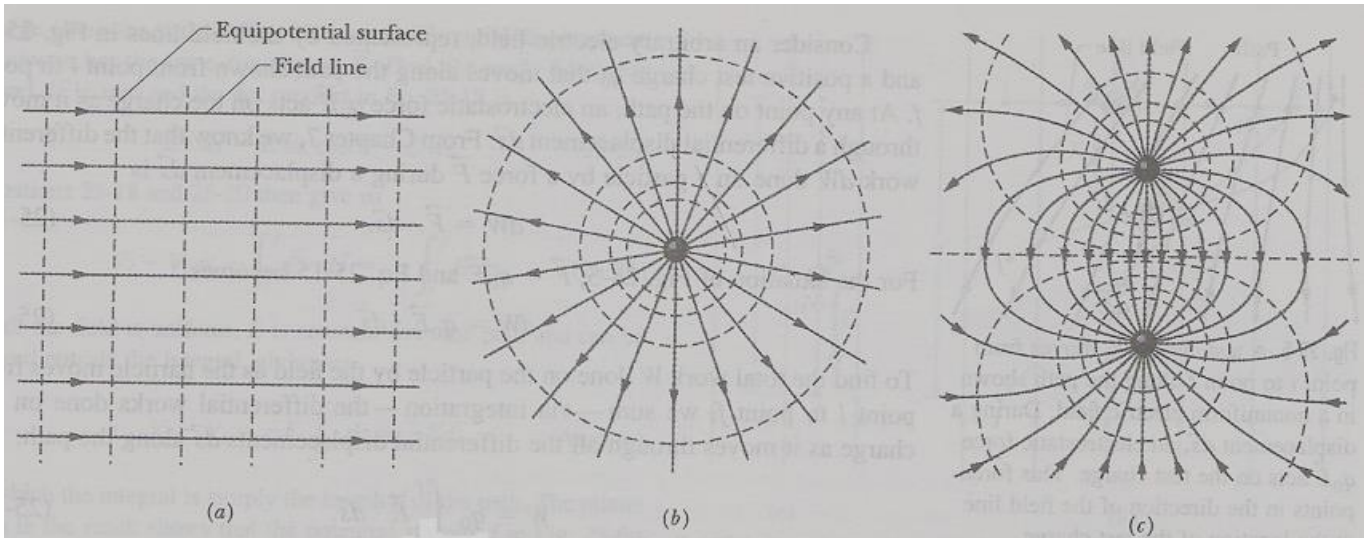
The work done by the electric field to move a charge q from a point with potential φ_i to a point with potential φ_f is:

$$W = q(\varphi_f - \varphi_i) = q\Delta\varphi \tag{II.17}$$

II.5. EQUIPOTENTIAL SURFACES

Adjacent points that have the same potential form an ***equipotential surface***. No net work W is done on a charge by the electric field when the charge moves between two points on the same equipotential surface. Fig.II.6 illustrates equipotential surfaces for some charge configurations

As we can see from this figure electric field lines are always perpendicular to equipotential surfaces.



Electric field lines (purple) and cross sections of equipotential surfaces (gold) for (*a*) a uniform field, (*b*) the field of a point charge, and (*c*) the field of an electric dipole.

Fig.II.6. Some equipotential surface configurations

II.6. CALCULATING THE POTENTIAL FROM THE FIELD

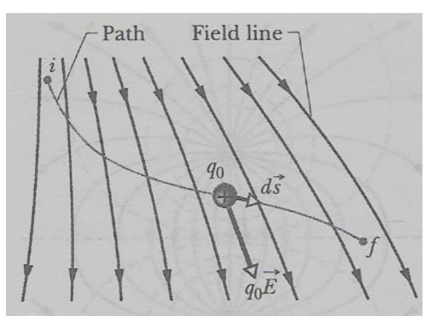
A charged particle moves from point i to point f . At any point on the point on the path, an electric force $q_0 E$ acts on the charge as it moves through a differential displacement $d\ell$ as shown in Fig.II.7:

$$dW = \bar{F} d\bar{\ell} = q_0 \bar{E} d\bar{\ell}$$

The total work W done on the particle by the field as the particle moves from point i to point f :

$$W = q_0 \int_i^f \bar{E} d\bar{\ell} \qquad \text{(II.17)}$$

And according to (II.17) we can write (II.12) as: $\varphi_f - \varphi_i = -\int_i^f \bar{E} d\bar{\ell}$



A test charge q_0 moves from point i to point f along the path shown in a nonuniform electric field. During a displacement $d\vec{s}$, an electrostatic force $q_0\vec{E}$ acts on the test charge. This force points in the direction of the field line at the location of the test charge.

Fig.II.7. Work done by an electrostatic field

(II.18)

II.7 POTENTIAL CREATED BY A POINT CHARGE

Consider a point P at a distance R from a fixed charged particle q. Imagine a test charge q_0 moving from P to infinity as in Fig.II.8. The path does not matter so let us choose the radial path from P to infinity. Here $\overline{F} \uparrow\uparrow d\overline{\ell}$ at any point:

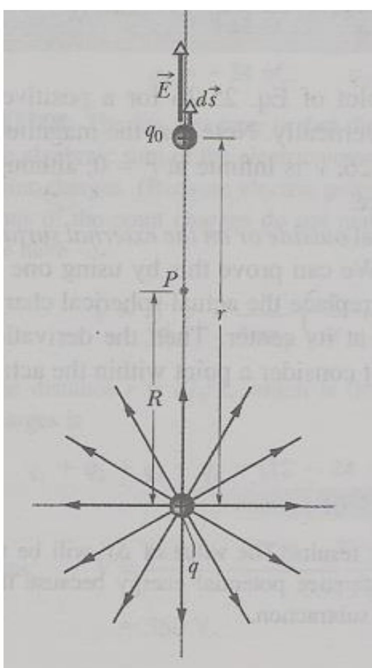
$$\overline{E}d\overline{\ell} = E\cos\theta dl\,, \qquad d\ell \equiv dr \qquad \rightarrow \qquad \varphi_f - \varphi_i = -\int_i^f \overline{E}d\overline{r}$$

If we set $\varphi_f = 0$ at ∞ and $\varphi_i = \varphi$ at R:

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2} \qquad \rightarrow \qquad 0 - \varphi = -\frac{q}{4\pi\varepsilon_0}\int_R^\infty \frac{1}{r^2}dr = \frac{q}{4\pi\varepsilon_0}\frac{1}{r}\bigg|_R^\infty = -\frac{1}{4\pi\varepsilon_0}\frac{q}{R} \qquad \rightarrow$$

$$\varphi = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \qquad \text{(II.19)}$$

A positively charged particle produces a positive electric potential. A negatively charged particle produces a negative electric potential.



The positive point charge q produces an electric field $\vec{E}$ and an electric potential V at point P. We find the potential by moving a test charge q_0 from P to infinity. The test charge is shown at distance r from the point charge, during differential displacement $d\vec{s}$.

Fig.II.8. Work done to bring a test charge from ∞ to a point P

II.8. POTENTIAL CREATED BY A GROUP OF POINT CHARGES

The ***superposition principle*** applies for the electric potential: ***the net potential produced by a system of point charges equal the algebraic sum of the potentials produced by each point charge separately***

$$\varphi = \sum_{i=1}^{n} \varphi_i = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{n} \frac{q_i}{r_i} \qquad \text{(II.20)}$$

This principle is a direct consequence of the principle of superposition of electrostatic force and the fact that work and energy are additive quantities

II.9. POTENTIAL DUE TO A CONTINUOUS CHARGE DISTRIBUTION

Another consequence is that when a charge distribution q is continuous as on a uniformly charged thin rod or disk, we choose a differential element of charge dq, determine the potential it creates at point P and integrate over the entire charge distribution

Let us take the zero of potential to be at infinity. We treat the element of charge dq as a point charge:

$$d\varphi = \frac{1}{4\pi\varepsilon_0} \frac{dq}{r},$$

where r is the distance between P and dq.

To find the total potential φ at P, we integrate to sum the potentials due to all the charge elements up:

$$\varphi = \int d\varphi = \frac{1}{4\pi\varepsilon_0} \int \frac{dq}{r} \qquad \text{(II.21)}$$

II.10. ELECTRIC POTENTIAL ENERGY OF A SYSTEM OF POINT CHARGES

The electric potential energy of a system of fixed charges is equal to the work that must be done by an external agent to assemble the system, bringing each charge in from an infinite distance

Let us consider two charges q_1 and q_2 separated by a distance r. We mentally build the system by taking charges from infinity. To bring q_1 in from infinity to its position does not need work as there is no electrostatic field yet. To bring in q_2 we perform work $q_2\varphi$ where φ is the electric potential set up by q_1:

$$\varphi = \frac{1}{4\pi\varepsilon_0} \frac{q_1}{r}$$

The electric potential energy of the pair of point charges is:

$$U = W = q_2\varphi = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r} \qquad \text{(II.22)}$$

II.11. POTENTIAL OF A CHARGED ISOLATED CONDUCTOR

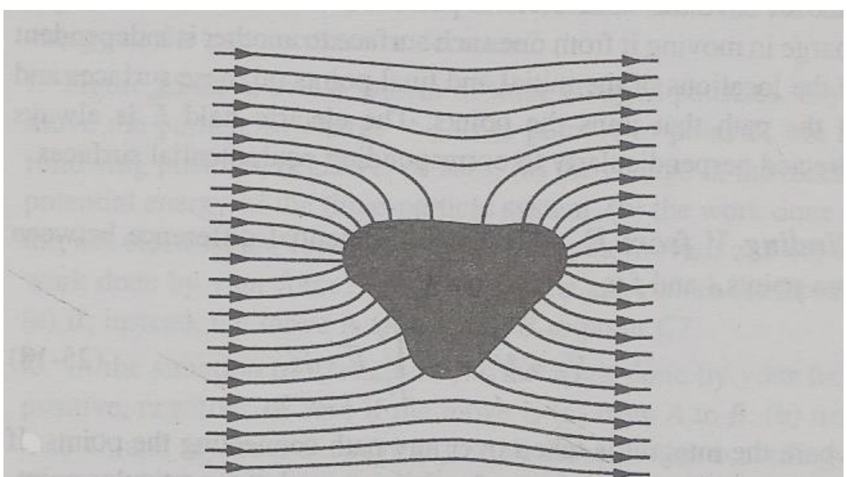
An excess charge placed on an isolated conductor will distribute itself on the surface of the conductor so that all points of the conductor, whether on the surface or inside, come to the same potential. This is due to the fact that like charges repel each other so they tend to place themselves as far as possible from each other therefore on the surface.

Therefore in electrostatics, we can make the following statements that are direct consequences of laws of electrostatics:

- A conductor must be an equipotential surface (and volume)
- Charges can only reside on the surface of the conductor and not inside the conductor
- Electric field intensity vector at all points on the surface of the conductor must be normal to the surface
- Electric field intensity inside an isolated conductor is zero: $E = 0$: there are no field lines within the conducting material
- A completely closed cavity inside a conductor cannot enclose any net charge
- If a charge is placed inside a cavity completely enclosed within a conductor, then an exactly equal and unlike charge must be induced on the outer surface
- Electric charges located outside a conductor cannot produce an electric field inside a completely enclosed cavity within the conductor
- If any point in space is at a maximum (or a minimum) of potential, then a positive (or negative) charge must present at that point.
- Field lines that start or stop on the surface of a conductor are perpendicular to the surface where they intersect it
- The density of field lines at the surface of the conductor is greatest at regions of smallest radius of curvature.

The last feature has an application in ***"lighting rod"*** in combination with the so called ***"Faraday's cage"***

Fig.II.9 illustrates distribution of electric field lines on a conductor placed in an external electric field and Fig.II.10 shows the distribution of the potential $\varphi(r)$ and electric field $E(r)$ and



An uncharged conductor is suspended in an external electric field. The free electrons in the conductor distribute themselves on the surface as shown, so as to reduce the net electric field inside the conductor to zero and make the net field at the surface perpendicular to the surface.

Fig.II.9. Uncharged conductor in an electric field

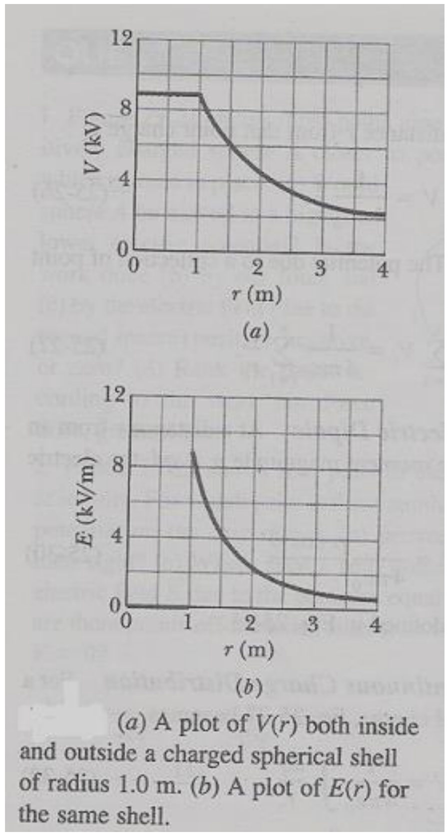


Fig.II.10. Electrical potential and electrical field due to a spherical conductor as function of distance

Practical activity II.3

Example II.4

Three point charges $q_1 = 1\mu C$, $q_2 = -2\mu C$, $q_3 = 3\mu C$ are fixed at positions shown in Fig.II.11. (a) What is the potential at point P at the corner of the rectangle? (b) How much work would be needed to bring a charge $q_4 = 2.5\mu C$ from infinity and to place it to P? What is the total potential energy of the system q_1, q_2, q_3?

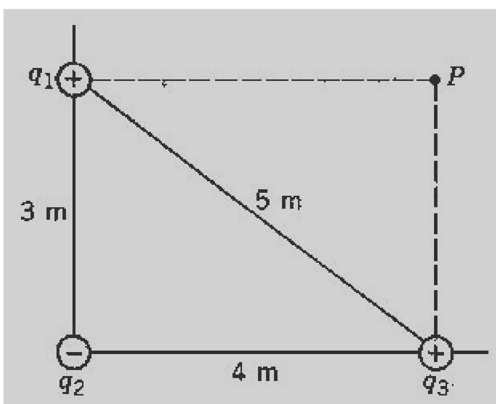


Fig.II.10. Example II.4

Solution

Numerical data: $q_1 = 1\mu C = 1X10^{-6}C$, $q_2 = -2\mu C = -2X10^{-6}C$, $q_3 = 3\mu C = 3X10^{-6}C$
$q_4 = 2.5\mu C = 2.5X10^{-6}C$, $d_{12} = 3m$ $d_{23} = 4m$, $d_{13} = 5m$, **(a)** $\varphi_P = ?$
(b) $W_{\infty \to P} = ?$ **(c)** $U = ?$

Solution: (a) The total potential at point P is the scalar sum of potentials: $\varphi_P = \varphi_1 + \varphi_2 + \varphi_3 = \dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{q_1}{r_1} + \dfrac{q_2}{r_2} + \dfrac{q_3}{r_3}\right)$.

With given values, $\varphi_1 = \dfrac{9X10^9(10^{-6})}{4} = 2.25X10^3 V$.

Similarly: $\varphi_2 = -3.6X10^3 V$ and $\varphi_3 = 9X10^3 V$.

The total potential at P is: $\varphi_P = (2.25 - 3.6 + 9)X10^3 V = 7.65X10^3 V$

(b) The external work is $W_{EXT} = -q_4(\varphi_f - \varphi_i)$.

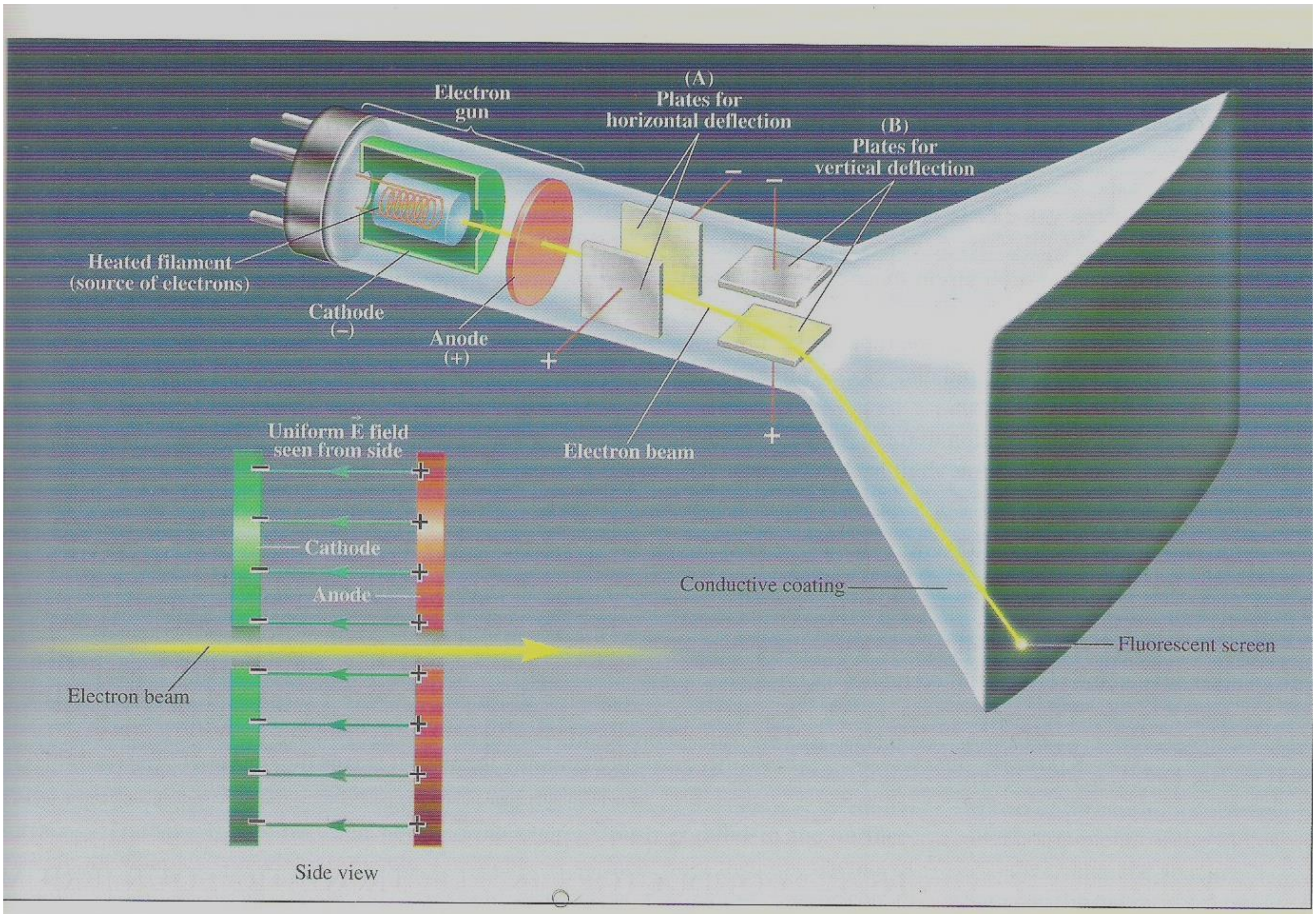
In this case $\varphi_i = \varphi_\infty = 0$ $\rightarrow$ $W_{EXT} = -q_4\varphi_4 = (2.5X10^{-6})(7.65X10^6) = 0.19V$

(c) The total potential energy of the three charges is the scalar sum: $U = U_{12} + U_{13} + U_{23} = \dfrac{1}{4\pi\varepsilon_0}\left(\dfrac{q_1q_2}{r_{12}} + \dfrac{q_1q_3}{r_{13}} + \dfrac{q_2q_3}{r_{23}}\right)$. We find for example $U_{12} = \dfrac{(9X10^9)(1.0X10^{-6})(-2X10^{-6})}{3} = -6X10^{-3} J$

Similarly: $U_{13} = +5.4X10^{-3} J$ and $U_{23} = -13.5X10^{-3} J$. The total potential energy is: $U = -1.41X10^{-3} J$

II.12. ONE VERY IMPORTANT APPLICATION OF ELECTROSTATICS: A CRT

Fig.II.11 beautifully illustrates one of the most important applications of electrostatics the ***Cathode Ray Tube*** (CRT) as used in an oscilloscope, ***CRO*** or ***Cathode Ray Oscilloscope***



In a cathode ray tube (CRT), electrons are accelerated to high speeds by an electric field between the athode and anode. This CRT, used in an oscilloscope, has two pairs of parallel plates that are used to deflect the electron eam horizontally and vertically.

Fig.II.11. A CRT used in a CRO

A cathode ray tube (CRT) is used to accelerate electrons in electronic displays: (old) TV sets, (old) computer monitors, (old) oscilloscopes and other devices such as X-ray tubes. (old) means that the CRT technology has become obsolete in devices like TV screens or computer monitors, but it has yet to find a replacement in devices like X–ray tubes

Summary

1. An electric charge influences its surroundings by creating an electric field. The electric field strength at a point is the force per unit positive charge on a test charge placed at that point:
$$\overline{E} = \frac{\overline{F}}{q_0}$$
2. The direction of E is that of the force on a positive charge
3. From the Coulomb's law we find the electric field created by a point charge Q as:

$$\overline{E} = k\frac{Q}{r^2}\overline{r}_0,$$

where the unit vector $\overline{r}_0$ has its origin at source charge Q.

4. When several charges are present, the total field is given by the principle of linear superposition

$$\overline{E} = \Sigma E_i$$

5. When the charge distribution is continuous, the field is obtained by integrating over the charge distribution

$$\overline{E} = k\int \frac{dQ}{r^2}\overline{r}_0$$

6. Electrical lines are a helpful visualization field patterns. They provide the following basic information:

(i) The strength or magnitude of the field: it is proportional to the number of lines that cross a unit area perpendicular to the lines

(ii) The direction of the field: its tangent to a line at a given point

7. Electric potential is a scalar that is related to work and potential energy is a conservative electric field. The change in potential ΔV in moving a unit positive charge from point A to point B is

$$V_A - V_B = \frac{W_{EXT}}{q} = \frac{\Delta U}{q},$$

where W_{EXT} is the work required to move the charge q from A to B at constant speed. $\Delta U = W_{EXT}$ is the associated change in potential energy.

8. Like the electric field, potential is function that depends on the source charges, not on any test charge. Only the changes in potential are significant, so one can arbitrarily choose a reference point and set for it $V_{REF} = 0$

9. Potential difference is related to the electric field:

$$V_B - V_A = -\int_A^B \overline{E}d\overline{\ell}$$

The integral does not depend on the path taken from A to B

10. In a uniform field, the change in potential is:

$$\Delta V = \left(\overline{E}.\overline{d}\right)$$

11. The potential at a distance r from a point charge Q is:

$$V(r) = k\frac{Q}{r}$$

It implies that $V = 0$ at $r = \infty$ and the sign of Q must be included

12. The potential due to a system of charges is given by the algebraic sum of the potentials due to individual charges. This is the principle of linear superposition of potential. The potential due to a continuous charge distribution is given by:

$$V(r) = \int \frac{kdq}{r}$$

13. The potential function may be represented by equipotential surfaces. In a two-dimensional plot the equipotentials are lines and the electric field lines are perpendicular to the equipotentials and points from higher to lower potential

14. The potential energy associated with a single charge at a point where the potential is due to other charges is V is:

    $U = qV$
15. Positive potential energy means that positive external work was required to bring the charges from infinity and place them at their given positions and negative potential energy means that positive external work must done to separate the charges
16. Once the scalar potential function V has been found the component of the electric field in the direction of the displacement dx may be found from:

    $$E_x = -\frac{dV}{dx}$$
17. For a homogeneous conductor in electrostatic equilibrium, the potential is the same at all points within and on the surface

Self-test II

1. An electric field is created by a set of fixed charges. When an external charge is introduced into the region, the field lines are modified. Should we use the original lines or the modified ones to determine the direction of the force on the new charge?
2. Electrostatic field lines begin on positive charges and end on negative charges! What happens to lines due to an isolated charges
3. Explain qualitatively why the field due to a an infinite sheet of charge is uniform
4. Name two fields that you encounter in everyday life that are (i) scalar (ii) vector
5. (i) If the potential is zero at a point, what can you say about the electric field at that point?

    (ii) If the field strength is at a point, what can you say about the potential function?
6. Points A and B have the same potential. In general would a net force be needed to move a charge from A to B? Would net work be required?
7. In a dry day, a spark between your finger and some object may involve several thousand volts. Why this is not dangerous when a mere $120V$ at wall outlet can be fatal?
8. Is the equation $\Delta V = \left(\overline{E}.\overline{d}\right)$ generally valid? Explain why or why not?
9. Is it possible to move a charge in an electric field without doing work? If so, how?
10. A metal shell of radius $R = 10cm$ is charged till its potential at surface is $120V$

    (i) What is the potential at the centre?

    (ii) What is the electric field at the centre?
11. As one follows a given field line in the direction of the field, does the potential increase, decrease or stay the same?
12. The electric filed strength inside a charged hollow cube is zero, but the gravitational field strength inside a hollow cubic mass distribution is not. Why the difference?

Problems II

1. A point charge $Q_0 = -4q$ is at $x = 0$ while the a second charge is at $x = 1m$. Besides infinity, where is the field strength zero given that the second charge is

    (i) $Q_1 = 9q$

    (ii) $Q_2 = -q$
2. A point charge $Q_0 = -5\mu C$ is at the origin. Find the electric field at the following points:

(i) $(2m, -1m)$

(ii) $(-2m, 3m)$

3. A point charge $Q_1 = -4\mu C$ is at $(2m, 1m)$ while $Q_2 = +15\mu C$ is at $(1m, 4m)$. Find the field strength at $(3m, 5m)$

4. An electron enters the region between two horizontal plates with initial velocity $v_{0X} = 2X10^6 \bar{i}\frac{m}{s}$ midway between two horizontal plates as shown in Fig.II.11

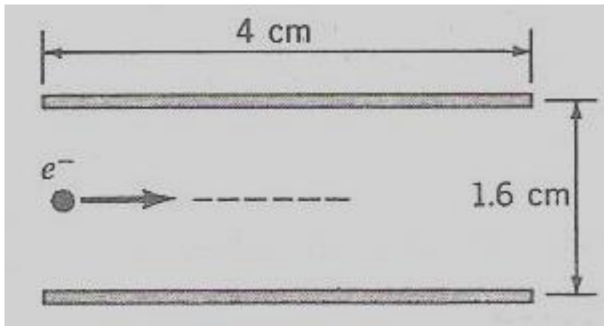


Fig.II.11. Problem 5

What is the magnitude of the maximum vertical electric filed such that the electron does not strike either plate?

5. A uniform field $E = -10^5 \bar{j}\frac{N}{C}$ exists between two plates of length $4cm$ as shown in Fig.II.12. A proton is fired at 30^0 to the x-axis with an initial speed of $v_0 = 8X10^5 \frac{m}{s}$. Find:

(i) Its vertical coordinate as it emerges from the plates

(ii) The angle at which it emerges

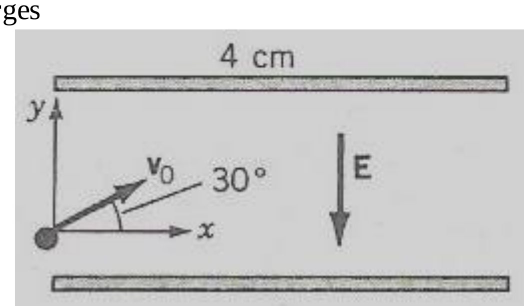


Fig.II.12 Problem 6

6. A point charge $Q_1 = 2\mu C$ is at a distance $d = 20cm$ from a uniformly charged infinite sheet with a surface charge density $\sigma = 20\frac{\mu C}{m^2}$.

(i) What is the force on the point

(ii) At what point is the resultant field strength zero?

7. A disc of radius $R = 4cm$ has a uniform charge density $\sigma = 5\frac{\mu C}{m^2}$. What is the field strength at a point along its central axis at a distance $d = 10cm$ from the centre?

8. A point charge $Q_1 = 10\mu C$ is placed at $(0, 3cm)$ and another point charge $Q_2 = -6\mu C$ is placed at $(4cm, 0)$

(i) What is the potential difference the origin and the point $(4cm, 3cm)$?

(ii) How much external work would it take to bring $q = 2\mu C$ point charge at constant speed from infinity to the origin

9. A point charge $Q_1 = 5\mu C$ is placed at the origin in Fig.II.13. Find the potential at the points (i) A, (ii) B and (iii) C

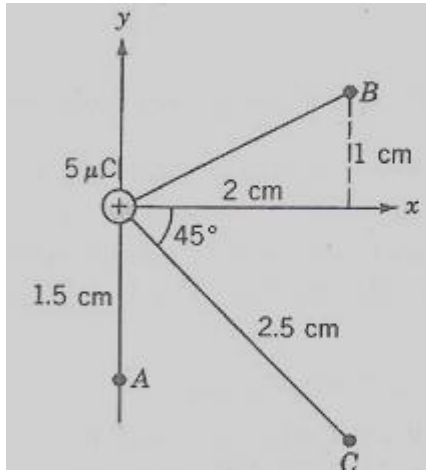


Fig.II.13. Problem 9

10. Two point charges are located as follows: $Q_1 = 3nC$ is at $\bar{r}_1 = 3\bar{i} - 2\bar{j} + \bar{k}$ and $Q_2 = -2nC$ is at $\bar{r}_2 = \bar{i} - 2\bar{j} + 6k, m$. Find:
    (i) The potential at origin
    (ii) The potential energy of A point charge $q = -5\mu C$ at origin
11. Suppose that a potential function is given by $V(x) = 3x^2 - 15x + 7, V$, where x is in metres. At what point is the electric field zero?
12. Two uniformly charged conducting spheres of radii $R_1 = 3cm$ and $R_2 = 7cm$ are connected by a wire and share a total charge $Q = 30nC$? What is the charge on each sphere?
13. A semicircular arc has a uniform linear charge density $\lambda = 2.2\frac{nC}{m}$. What is the potential at the centre?
14. When the electric field reaches $E = 3\frac{MV}{m}$, air breaks down (a spark is seen). What is the maximum potential possible for a metal sphere of radius $R = 40cm$
15. The speed of a proton moving antiparallel to the lines of an electric field is reduced from $v_1 = 2.4X10^6 \frac{m}{s}$ to $v_2 = 8X10^5 \frac{m}{s}$. What is the change in potential between the two points?
16. A rod of length L with a uniform linear charge density λ lies along the x axis. Find the potential at a distance y from the end, perpendicular to the rod as shown in Fig.II.14.

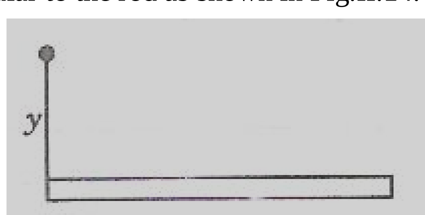


Fig.II.14 Problem 17

LECTURE III. CAPACITORS AND DIELECTRICS

III.1. LECTURE OUTLINE

When charge on a conductor increases its potential increases proportionally. The proportionality coefficient between charge and potential is called capacitance. It has been observed that the capacitance of isolated conductors is very small. To increase the capacitance so to store more charge, a system of two conductors placed very close to each other is used. Such a system is called a capacitor. Capacitors can be combined in series or in parallel. A charged capacitor stores electric energy. Capacitance of a capacitor can be increased if the space between the two conductors, called plates is filled by a dielectric. Because the dielectric becomes polarized this allows to store more charges at the same potential difference between the plates. Capacitors are used extensively in electrical and electronic circuits

III.1.1. SPECIFIC OBJECTIVES

At the end of this lecture you should be able to:

1. Define the capacitance of a conductor
2. Describe a capacitor and its applications
3. Compute the capacitance and energy stored in a plane- plate capacitor
4. Describe the effects of a dielectric on capacitance of a capacitor and the physical mechanism behind them
5. Solve problem related to combinations of capacitors and stored energy

III.2. CAPACITORS AND CAPACITANCE

Two conductors isolated electrically from each other and from the surroundings form a ***capacitor***. When a capacitor is charged, the conductors also called ***plates*** have ***equal*** but ***opposite charges*** of magnitude q as shown in Fig.III.1

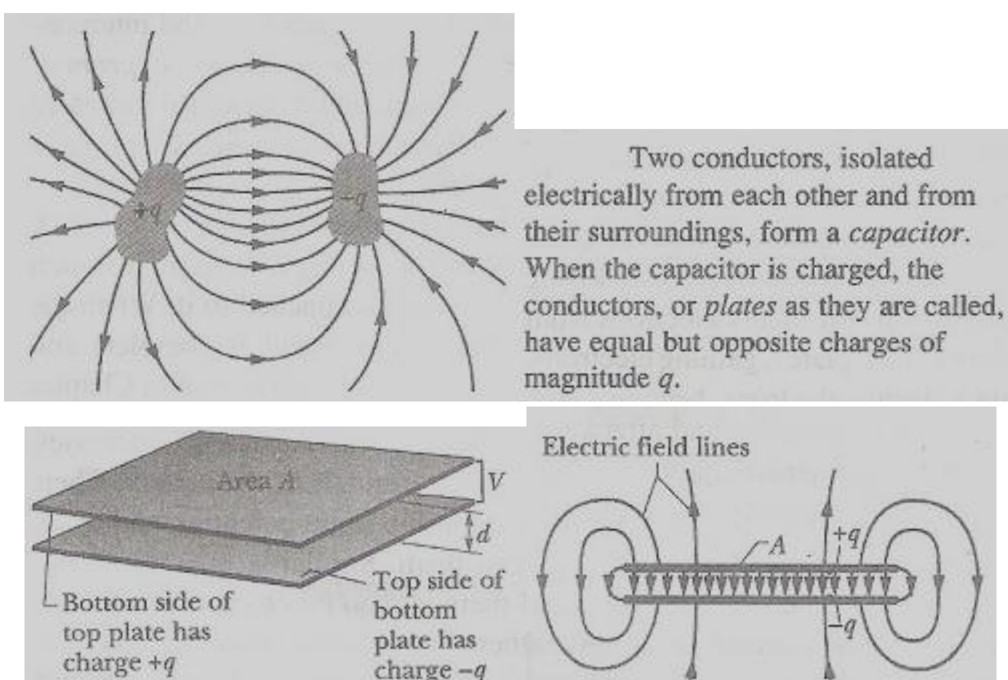


Fig.III.1. Two conductors placed close together carrying equal and opposite charge form a capacitor

Fig.III.2 shows some types of capacitors mounted on a printed electronic circuit.

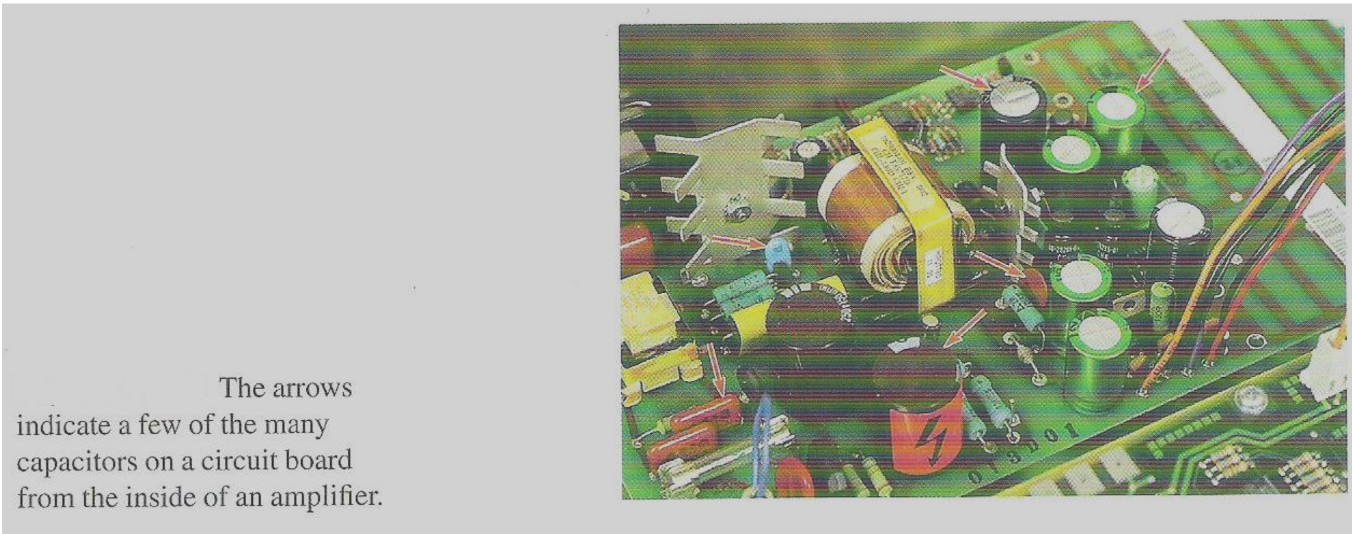


Fig.III.2. Some types of capacitors on an electronic printed board

The charge q and the potential difference V between the plates of a capacitor are proportional to each other:

$$q = CV \tag{III.1}$$

where the proportionality coefficient C is called ***capacitance***. Its value depends exclusively on the geometry of the plates and not on the charge or the potential difference.

The capacitance is a measure of how much charge must be on the plates to produce a certain potential difference: the greater the capacitance the more charge is required.

"***Capacitance is the charge required to increase the potential difference by*** $1 Volt$".

The ***SI unit of capacitance*** is ***Farad, F:***

$$[C] = 1F = \frac{1C}{1V} \tag{III.2}$$

The farad is a very large unit. Submultiples of the farad such as ***microfarad***, $1\mu F = 10^{-6} F$, ***nanofarad***, $1nF = 10^{-9} F$ and ***picofarad***, $1pF = 10^{-12} F$ are more convenient. One-farad-capacitors nowadays are available but they are used in special applications and are quite expensive.

The word ***Farad*** is derived from ***Faraday, Michael****, a giant in history of the development of electromagnetic theory and science in general

****Michael Faraday,*** FRS **(22 September 1791 – 25 August 1867)** was an ***English scientist*** who contributed to the fields of ***electromagnetism*** and ***electrochemistry***. His main discoveries include those of ***electromagnetic induction,***

diamagnetism and ***electrolysis***. Although Faraday received little formal education he was one of the most influential scientists in history, and historians of science refer to him as having been the best experimentalist in the history of science. It was by his research on the ***magnetic field*** around a conductor carrying a ***direct current*** that Faraday established the basis for the concept of the ***electromagnetic field*** in physics. Faraday also established that magnetism could affect rays of light and that there was an underlying relationship between the two phenomena. He similarly discovered the principle of ***electromagnetic induction, diamagnetism***, and the ***laws of electrolysis***. His inventions of electromagnetic rotary devices formed the foundation of ***electric motor*** technology, and it was largely due to his efforts that electricity became practical for use in technology.

One-farad capacitor manufactured by Sanyo is represented in Fig.III.3

Fig.III.3. One-farad capacitor manufactured by Sanyo

III.2.1. CHARGING A CAPACITOR

The experimental setup shown in Fig.III.4(a) can be used to charge a capacitor. It shows also the symbol for a capacitor.

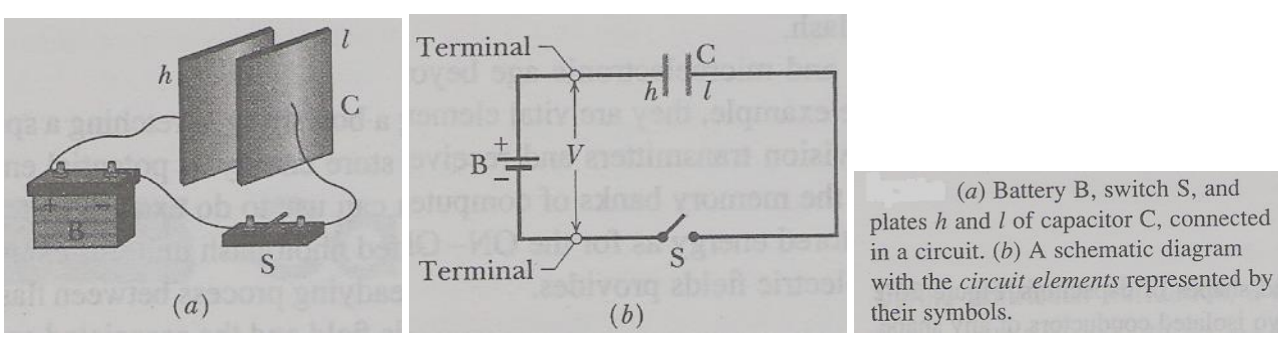


Fig.III.4. Charging a capacitor

III.2.2. CALCULATING THE CAPACITANCE

To calculate the capacitance C of a given geometry:

1. We assume a charge q on the plates
2. We calculate the electric field E between the plates
3. We calculate the potential difference V from E
4. Calculate C from expression (III.1)

Using the methodology above it can be proved that the following geometries have respective capacitances

1. Plane-plates capacitor

For a ***plane-plate*** or ***parallel-plate capacitor*** represented in Fig.III.5, the capacitance C is:

$$C = \frac{\varepsilon_0 A}{d} \qquad \text{(III.3)},$$

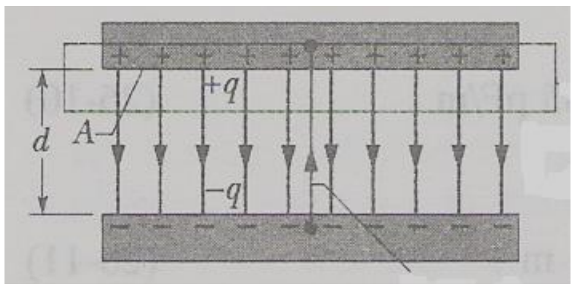


Fig.III.4. Charged plane-plate capacitor

where A the "***mutual" area of plates***, d the ***separation*** between the plates

Expression (III.3) can help to define SI unit for ε_0. The expression (I.9) gives:

$$\varepsilon_0 = \frac{1}{36\pi X10^9}\frac{C^2}{N.m^2} = 8.85X10^{-12}\frac{C^2}{N.m^2} = 8.85X10^{-12}\frac{F}{m} \qquad \text{(III.4)}$$

2. Cylindrical capacitor

For a cylindrical capacitor whose cross section is represented in Fig.III.6, the capacitance C is:

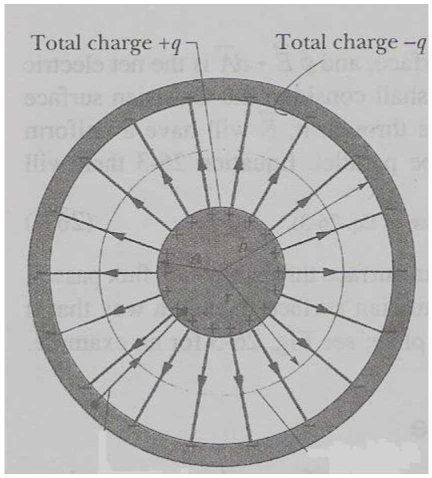


Fig.III.6. A cross-section of a cylindrical capacitor

$$C = 2\pi\varepsilon_0 \frac{L}{\ln\left(\frac{b}{a}\right)} \qquad \text{(III.5)},$$

where L is the length of capacitor, a and b are internal and external radii of the cylinder as represented in Fig.III.4

3. Spherical capacitor

$$C = 4\pi\varepsilon_0 \frac{ab}{b-a} \qquad \text{(III.6)},$$

where a and b are internal and external radii of the sphere.

4. An isolated sphere

For an isolated sphere we can use expression (III.6) as: $C = 4\pi\varepsilon_0 \frac{a}{1-\frac{a}{b}}$

and let $b \to \infty$ and substitute a for R we get:

$$C = 4\pi\varepsilon_0 R \qquad \text{(III.7)}$$

Expressions (III.3)-(III.7) show that capacity is proportional to the size of the capacitor, and it is logical because the bigger is the conductor the more charge it can hold without its potential rising too much.

III.4. COMBINATION OF CAPACITORS

When there is a combination of capacitors in a circuit, we can replace it with an equivalent capacitor that has the same capacitance C as the combination

III.4.1. CAPACITORS IN PARALLEL

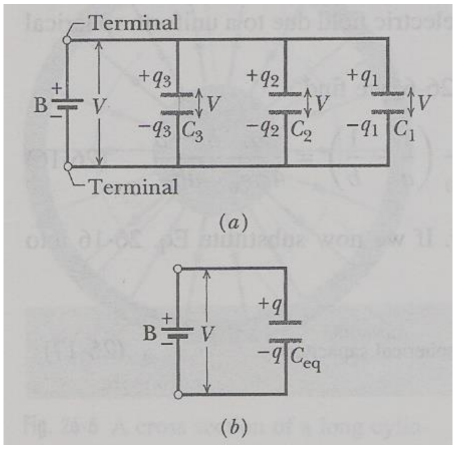
When a potential difference V is applied cross several capacitors connected in parallel, that potential difference is applied across each capacitor as shown in Fig.III.7

The total charge q stored on the capacitors is the sum of the charge stored on all the capacitors

→ Capacitors connected in parallel can be replaced with an equivalent capacitor that has the same total charge q and the same potential difference V as the actual capacitors. To derive an expression for C_{eq} :

$q_1 = C_1 V$, $\quad q_2 = C_2 V$ and $q_3 = C_3 V$

The total charge on the parallel combination is then: $\quad q = q_1 + q_2 + q_3 = (C_1 + C_2 + C_3)V$



(a) Three capacitors connected in parallel to battery B. The battery maintains potential difference V across its terminals and thus across *each* capacitor. (b) The equivalent capacitor, with capacitance C_{eq}, replaces the parallel combination.

Fig.III.7. Capacitors in parallel

The equivalent capacitance is: $\quad C_{eq} = \dfrac{q}{V} = (C_1 + C_2 + C_3)$

For generality to n capacitors in parallel:

$$C_{eq} = \sum_{i=1}^{n} C_i \qquad \text{(III.8)}$$

Expression (III.8) means that the equivalent ***capacitance*** of capacitors connected in ***parallel*** equals the ***sum*** of individual capacitances

III.4.2. CAPACITORS IN SERIES

A combination of capacitors in series is given in Fig.III.6. When a potential difference V is connected across the combination, the capacitors have identical charge q and the sum of the individual potential differences equals the applied potential difference V .

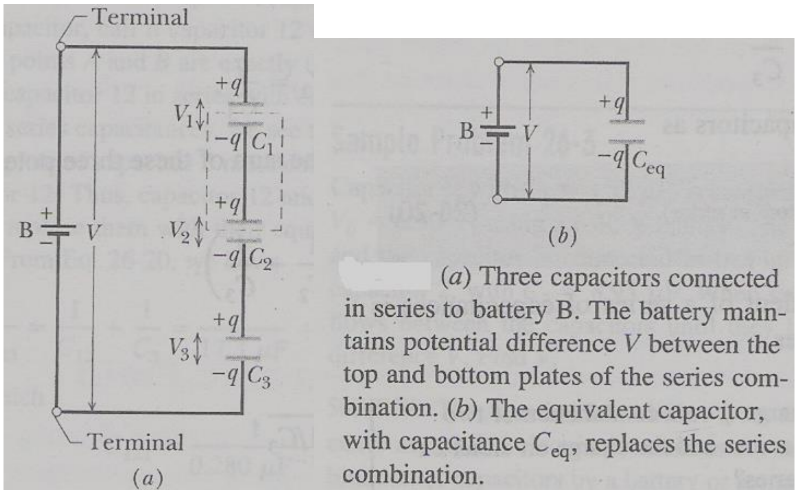
The combination can be replaced with an equivalent capacitor that has the same charge q and the same total potential difference V as the actual series capacitors:

$$V_1 = \frac{q}{C_1}, V_2 = \frac{q}{C_2} \text{ and } V_3 = \frac{q}{C_3} \text{ but } V = V_1 + V_2 + V_3 = q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$

$$\rightarrow C_{eq} = \frac{q}{V} = \frac{q}{q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)} \qquad \rightarrow \quad \frac{1}{C_{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)$$

For n capacitors in series:

$$\frac{1}{C_{eq}} = \sum_{i=1}^{n} \frac{1}{C_i} \qquad \text{(III.9)}$$



(a) Three capacitors connected in series to battery B. The battery maintains potential difference V between the top and bottom plates of the series combination. (b) The equivalent capacitor, with capacitance C_{eq}, replaces the series combination.

Fig.III.6. Capacitors in series

Practical activity III.1

Example III.1

For the circuit in Fig.III.7 (a), find: (a) the equivalent capacitance, (b) the charge and potential difference for each capacitor?

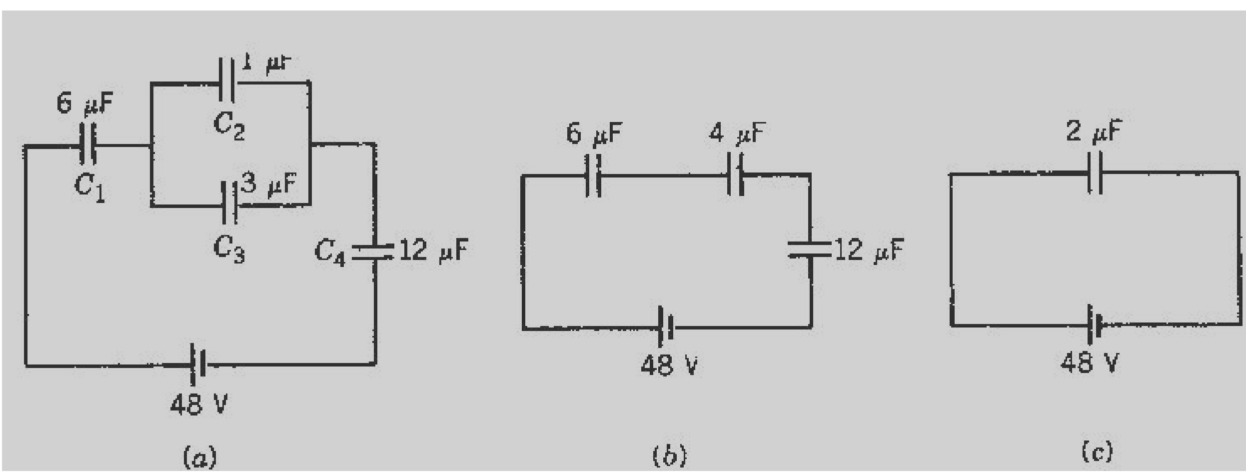


Fig.III.7 Example III.1

Solution

When dealing with large combinations of capacitors, one should start by simplifying the combination with smallest number of elements in series or parallel.

(a) We start with C_2 and C_3 which are in parallel and are equivalent to a $4\mu F$ capacitor as shown in Fig.III.7b where it is in series with C1 and C4. The total equivalent capacitance in Fig.III.6c is given by $\frac{1}{C_{eq}} = \frac{1}{6} + \frac{1}{4} + \frac{1}{12} = \frac{1}{2} \rightarrow C_{eq} = 2\mu F$

(b) Since capacitors in series have the same charge, $Q_1 = Q_2 + Q_3 = Q_4$. This is the also the charge on the equivalent capacitance, $Q = C_{eq}V = (2\mu C)(48V) = 96\mu C$ Thus $Q_1 = Q_4 = 96\mu C$

(c) To find the charges on the other two capacitors, we need their common potential difference: $V_2 = V_3$. We first find the potential across C1 and C4: $V_1 = \frac{Q_1}{C_1} = \frac{96\mu C}{6\mu F} = 16V$ $V_1 = \frac{Q_4}{C_4} = \frac{96\mu C}{12\mu F} = 8V$. $\rightarrow V_2 = V_3 = V_{BAT} - (V_1 + V_4) = 24V$. Finally charges on C2 and C3: $Q_2 = C_2V_2 = (1\mu F)(24V) = 24\mu C$ and $Q_3 = C_3V_3 = (3\mu F)(24V) = 72\mu C$

III.5. ENERGY STORED IN AN ELECTRIC FIELD

Suppose that a charge $q^/$ has been transferred from one plate of a capacitor to the other. The potential difference $V^/$ between the plates at this instant is $V^/ = \frac{q^/}{C}$. If an extra increment of charge $dq^/$ is transferred, the increment of work required: $dW = V^/dq^/ = \frac{q^/}{C}dq^/$

The work required to bring the total capacitor charge to a final value q is:

$$W = \int dW = \frac{1}{C}\int_0^q q^/dq^/ = \frac{q^2}{2C}$$

This work is stored as potential energy U in the capacitor:

$$U = \frac{q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}qV \qquad \text{(III.10)}$$

III.5.1. ENERGY DENSITY

Energy density is the energy stored in capacitor per unit volume:

$$u_E = \frac{U_E}{Vol} \qquad \text{(III.11)},$$

Where Vol is the volume of the space between the plates where the electric field is confined. For a plane-plate capacitor:

$$u_E = \frac{U_E}{Vol} = \frac{1}{2}\frac{CV^2}{Ad} = \frac{1}{2}\frac{\varepsilon_0\frac{A}{d}(Ed)^2}{Ad} = \frac{1}{2}\varepsilon_0E^2 \qquad \text{(III.12)},$$

where E is the electric field that is uniform in space between the plates. Expression (III.12) shows that energy stored in a capacitor resides actually in the electric field not in charges or the potential difference

III.6. FOR FURTHER READING: APPLICATIONS OF CAPACITORS

A capacitor can store electric energy when disconnected from its charging circuit, so it can be used like a temporary battery. Capacitors are commonly used in electronic devices to maintain power supply while batteries are being changed. (This prevents loss of information in volatile memory.)

Conventional electrostatic capacitors provide less than 360 joules per kilogram of energy density, while capacitors using developing technology can provide more than 2.52 kilojoules per kilogram.

In car audio systems, large capacitors store energy for the amplifier to use on demand.

An ***uninterruptible power supply (UPS)*** can be equipped with maintenance-free capacitors to extend service life.

III.6.1. PULSED POWER AND WEAPONS

Groups of large, specially constructed, low-inductance high-voltage capacitors ***(capacitor banks)*** are used to supply huge pulses of current for many pulsed power applications. These include ***electromagnetic forming***, ***Marx generators, pulsed lasers*** (especially ***TEA lasers***), ***pulse forming networks, fusion research, and particle accelerators.***

Large capacitor banks ***(reservoirs)*** are used as energy sources for the exploding- ***bridgewire detonators*** or ***slapper detonators*** in nuclear weapons and other specialty weapons. Experimental work is under way using banks of capacitors as power sources for ***electromagnetic armour***s and ***electromagnetic railguns*** or ***coilguns***.

III.6.2. POWER CONDITIONING

Reservoir capacitors are used in power supplies where they smooth the output of a ***full half wave rectifier***. They can also be used in charge pump circuits as the energy storage element in the generation of higher voltages than the input voltage.

Capacitors are connected in parallel with the power circuits of most electronic devices and larger systems (such as factories) to shunt away and conceal current fluctuations from the primary power source to provide a "clean" power supply for signal or control circuits. Audio equipment, for example, uses several capacitors in this way, to shunt away power line hum before it gets into the signal circuitry. The capacitors act as a local reserve for the DC power source, and bypass AC currents from the power supply. This is used in car audio applications, when a stiffening capacitor compensates for the inductance and resistance of the leads to the lead-acid car battery.

III.6.3. POWER FACTOR CORRECTION

In electric power distribution, capacitors are used for ***power factor correction***. Such capacitors often come as three capacitors connected as a three phase load. Usually, the values of these capacitors are given not in farads but rather as a reactive power in ***volt-amperes reactive (var)***. The purpose is to counteract inductive loading from devices like electric motors and transmission lines to make the load appear to be mostly resistive. Individual motor or lamp loads may have capacitors for power factor correction, or larger sets of capacitors (usually with automatic switching devices) may be installed at a load center within a building or in a large utility substation.

III.6.4. SUPPRESSION AND COUPLING

Signal coupling

Because capacitors pass AC but block DC signals (when charged up to the applied dc voltage), they are often used to separate the AC and DC components of a signal. This method is known as *AC coupling* or "capacitive coupling". Here, a large value of capacitance, whose value need not be accurately controlled, but whose reactance is small at the signal frequency, is employed.

Decoupling

A decoupling capacitor is a capacitor used to decouple one part of a circuit from another. Noise caused by other circuit elements is shunted through the capacitor, reducing the effect they have on the rest of the circuit. It is most commonly used between the power supply and ground. An alternative name is ***bypass capacitor*** as it is used to bypass the power supply or other high impedance component of a circuit.

Noise filters and snubbers

When an inductive circuit is opened, the current through the inductance collapses quickly, creating a large voltage across the open circuit of the switch or relay. If the inductance is large enough, the energy will generate an electric spark, causing the contact points to oxidize, deteriorate, or sometimes weld together, or destroying a solid-state switch. A ***snubber capacitor*** across the newly opened circuit creates a path for this impulse to bypass the contact points, thereby preserving their life; these were commonly found in contact breaker ignition systems, for instance. Similarly, in smaller scale circuits, the spark may not be enough to damage the switch but will still radiate undesirable radio frequency interference (RFI), which a **filter** capacitor absorbs. Snubber capacitors are usually employed with a low-value resistor in series, to dissipate energy and minimize RFI. Such resistor-capacitor combinations are available in a single package.

Capacitors are also used in parallel to interrupt units of a ***high-voltage circuit breaker*** in order to equally distribute the voltage between these units. In this case they are called grading capacitors.

In schematic diagrams, a capacitor used primarily for DC charge storage is often drawn vertically in circuit diagrams with the lower, more negative, plate drawn as an arc. The straight plate indicates the positive terminal of the device, if it is polarized (see electrolytic capacitor).

III.6.5. MOTOR STARTERS

In ***single phase squirrel cage motors***, the primary winding within the motor housing is not capable of starting a rotational motion on the rotor, but is capable of sustaining one. To start the motor, a secondary winding is used in series with a non-polarized ***starting capacitor*** to introduce a lag in the sinusoidal current through the starting winding. When the secondary winding is placed at an angle with respect to the primary winding, a rotating electric field is created. The force of the rotational field is not constant, but is sufficient to start the rotor spinning. When the rotor comes close to operating speed, a centrifugal switch (or current-sensitive relay in series with the main winding) disconnects the capacitor. The start capacitor is typically mounted to the side of the motor housing. These are called capacitor-start motors, and have relatively high starting torque.

There are also capacitor-run induction motors which have a permanently connected phase-shifting capacitor in series with a second winding. The motor is much like a two-phase induction motor.

Motor-starting capacitors are typically non-polarized electrolytic types, while running capacitors are conventional paper or plastic film dielectric types.

III.6.6. SIGNAL PROCESSING

The energy stored in a capacitor can be used to represent information, either in binary form, as in DRAMs, or in analogue form, as in analog sampled filters and CCDs. Capacitors can be used in analog circuits as components of integrators or more complex filters and in negative feedback loop stabilization. Signal processing circuits also use capacitors to integrate a current signal.

Tuned circuits

Capacitors and inductors are applied together in tuned circuits to select information in particular frequency bands. For example, radio receivers rely on variable capacitors to tune the station frequency. Speakers use passive analog crossovers, and analog equalizers use capacitors to select different audio bands.

The resonant frequency f_0 of a tuned circuit is a function of the inductance (L) and capacitance (C) in series, and is given by:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

where L is in Henry and C is in Farad.

III.6.7. SENSING

Most capacitors are designed to maintain a fixed physical structure. However, various factors can change the structure of the capacitor; the resulting change in capacitance can be used to ***sense*** those factors.

Varying the characteristics of the dielectric

The effects of varying the characteristics of the dielectric can also be used for sensing and measurement. Capacitors with an exposed and porous dielectric can be used to measure humidity in air. Capacitors are used to accurately measure the fuel level in airplanes; as the fuel covers more of a pair of plates, the circuit capacitance increases.

Changing the distance between the plates

A capacitor with a flexible plate can be used to measure strain or pressure or weight.

Industrial pressure transmitters used for process control use pressure-sensing diaphragms, which form a capacitor plate of an oscillator circuit. Capacitors are used as the sensor in condenser microphones, where one plate is moved by air pressure, relative to the fixed position of the other plate. Some accelerometers use ***microelectromechanical systems (MEMS)*** capacitors etched on a chip to measure the magnitude and direction of the acceleration vector. They are used to detect changes in acceleration, e.g. as tilt sensors or to detect free fall, as sensors triggering airbag deployment, and in many other applications. Some fingerprint sensors use capacitors.

Additionally, a user can adjust the pitch of ***a theremin*** musical instrument by moving his hand since this changes the effective capacitance between the user's hand and the antenna.

Changing the effective area of the plates

Capacitive touch switches are now used on many consumer electronic products.

Capacitor microphone

A capacitor microphone is represented in Fig.III.8

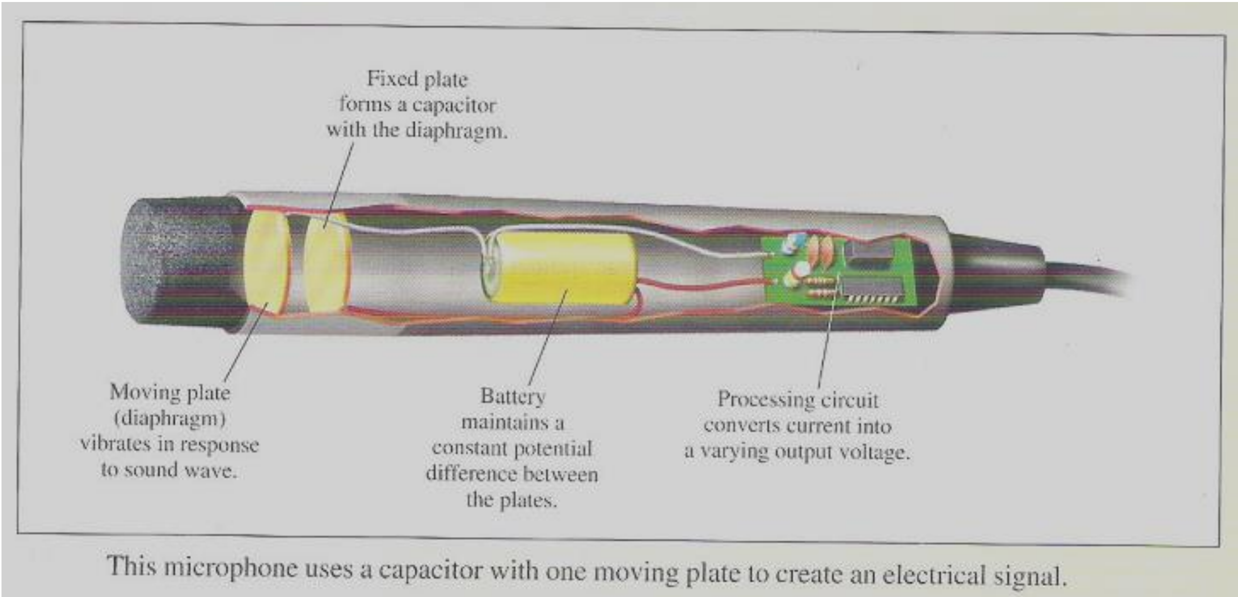


Fig.III.8. A capacitor microphone

Practical activity III.2

Example III.2

A $16\mu F$ capacitor is charged to $100V$. After being disconnected, it is immediately connected in parallel with uncharged $4\mu F$ capacitor. Determine (a) the potential difference across the combination and (b) the electrostatic energies before and after the capacitors are connected in parallel

Solution

Numerical data: $C_1 = 16\mu F = 1.6X10^{-5}F$ $C_2 = 4\mu F = 4X10^{-6}F$,
$V_{1i} = 100V$ $V_{1f} = ?$ $V_{2f} = ?$ $U_{1f} = ?$ $U_{2f} = ?$

Solution: (i) Before joining: Charge on $C_1 : Q_1 = C_1V_1 = (1.6X10^{-5})(100) = 1.6X10^{-3}C$

Energy stored: $U_1 = \frac{1}{2}C_1V_1^2 = \frac{1}{2}(1.6X10^{-5})(100)^2 = 0.08J$

(ii) After joining: We apply law of conservation of electric charge. When capacitors are in parallel the equivalent capacitance: $C_{eq} = C_1 + C_2 = 20\mu F$.

The Pd across the parallel connection is: $V = \frac{Q}{C_T} = \frac{1.6X10^{-3}}{20X10^{-6}} = 80V$.

Energy stored: $U_1 = \frac{1}{2}C_{eq}V^2 = \frac{1}{2}(2.0X10^{-5})(80)^2 = 0.064J$

We see that there is loss of energy due to heat dissipated in wires when charges distribute themselves between the plates

IV.7. CAPACITOR WITH A DIELECTRIC

If you fill the space between the plates of a capacitor with a dielectric, an insulating material like mineral oil or plastic, what happens with the capacitance?
This question was extensively investigated by ***M. Faraday***. He found that the capacitance increased by a numeral factor $\varepsilon \geq 1$ which he called the ***dielectric constant*** of the insulating material.
Another effect of the introduction of a dielectric is to limit the potential difference that can be applied between the plates to a certain value $V_{\max}$, the ***breakdown potentia***l. If this potential is substantially exceeded, the material will breakdown and forms a conducting path between the plates and the capacitor is destroyed.

Every dielectric material has a ***characteristic dielectric strength*** which is the maximum value of the electric field that it can tolerate without breakdown

The ***dielectric strength*** is measured in $\frac{kV}{mm}$

The next Table IV.1 gives the value of dielectric constant for some common materials

Table IV.1 Dielectric constant and dielectric strength for some common materials

Material	**Dielectric constant**	**Dielectric strength,** $k\frac{V}{mm}$	**Material**	**Dielectric constant**	**Dielectric strength,** $k\frac{V}{mm}$
Air	1.00054	3	**Silicon**	12	
Kerosene	2		**Germanium**	16	
Paraffin	2		**Ethanol**	25	
Polystyrene	2.6	24	**Water** 20^0	80.4	
Paper	3.5	16	**Titania ceramic**	130	
Transformer oil	4.5		**Strontium titanate**	310	
Pyrex	4.7		**Barium titanate**	1200	
Mica	5.4				
Glass	7.0				

III.8. LAWS OF ELECTROSTATICS IN DIELECTRICS

In a region completely filled by a dielectric material of dielectric constant ε all electrostatic equations containing the permittivity constant ε_0 are to be modified by replacing ε_0 with $\varepsilon_0\varepsilon$.
Thus a point charge inside a dielectric produces an electric field:

$$E = \frac{1}{4\pi\varepsilon_0\varepsilon}\frac{q}{r^2} \tag{III.13}$$

This equation shows that for a fixed distribution of charges the effect of the dielectric is to weaken the electric field that would be otherwise present.

IV.8. DIELECTRICS: AN ATOMIC POINT OF VIEW

In atomic and molecular terms, depending on the nature of the molecules two situations can happen:

Polar molecules

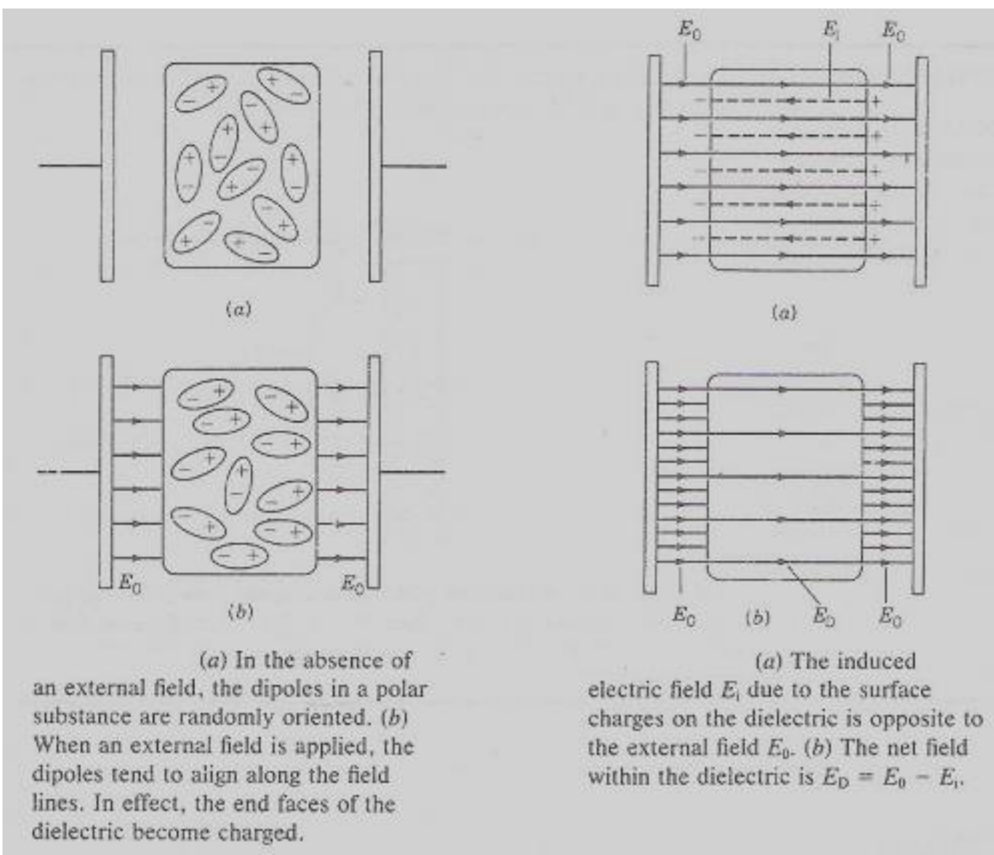
The molecules of some dielectrics like water have a permanent ***electric dipole moment***. The electric dipoles tend to line up opposite the external electric field. Because of the thermal agitation, this alignment can not be complete but it becomes more complete as the magnitude of the electric field is increased. The alignment of the dipoles produces an electric field that is directed opposite the applied field and smaller in magnitude

Non polar molecules

Regardless whether they have permanent electric dipole moments or not, molecules acquire dipole moments when placed in an external electric field and they become polarized. This has the same effect as for polar molecules as illustrated in Fig.III.8

Thus, if E_0 is the external electric field and $E^{/}$ the electric field produced by induced charges in the dielectric, the resultant or effective field in the dielectric is:

$$E = E_0 - E^{/} \qquad \text{(III.13)}$$



(a) In the absence of an external field, the dipoles in a polar substance are randomly oriented. (b) When an external field is applied, the dipoles tend to align along the field lines. In effect, the end faces of the dielectric become charged.

(a) The induced electric field E_i due to the surface charges on the dielectric is opposite to the external field E_0. (b) The net field within the dielectric is $E_D = E_0 - E_i$.

Fig.III.8. Dielectric in an electric field

Expression (III.12) for the stored energy density in a capacitor with a dielectric becomes:

$$u_E = \frac{1}{2}\varepsilon_0 \varepsilon E^2 \tag{III.14}$$

This shows that the insertion of a dielectric into a capacitor increases ε times the energy stored compared to an air-filled capacitor, if other parameters remain the same. This is one of the advantages of using dielectrics in manufacture of capacitors.

Summary

A ***capacitor*** is a device that stores charge and electrical energy. It consists of two conducting plates separated by an insulator. If the plates carry charges $\pm Q$ and have a potential difference V then the ***capacitance*** C is defined as:

$$C = \frac{Q}{V}$$

Capacitance depends on the size and shape of the plates and the material between them. It does not depend on Q or V individually

For plane-plate or parallel-plate capacitors (they are the most commonly used): $C = \varepsilon_0 \frac{A}{d}$,

where A is the area of the plates and d the separation distance between the plates.

When capacitors are connected in series the equivalent capacitance is given by:

$$\frac{1}{C_{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} + ... + \frac{1}{C_3} \right) = \sum_{i=1}^{N} \frac{1}{C_i}$$

When they are connected in parallel, the equivalent capacitance is given by: $C_{eq} = \sum_{i=1}^{n} C_i$

To charge a capacitor the work done by the external agent is stored as electrical potential energy:

$$U_E = \frac{1}{2} QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} CV^2$$

The energy of a capacitor is actually stored in the electric field that is created between the plates. The ***energy density*** of an electric field is given by: $u_E = \frac{1}{2}\varepsilon_0 \varepsilon E^2$

Suppose a capacitor has capacitance C_0 when there is no material between the plates (an air-filled capacitor). When a dielectric is inserted to completely fill the space between the plates, the capacitance increases to:

$$C = \varepsilon C_0$$

where ε is called the ***dielectric constant*** of the material

Self-test III

1. When a battery is connected across a capacitor are the charges on the plates always equal and opposite, even for plates of different size?

2. Would two conductors have capacitance even if they did not have equal and opposite charges? Does a single conductor have capacitance?

3. Given a battery, how would you connect two capacitors, in series or in parallel , for them to store the greater:

   (i) Total charge

   (ii) Total energy

4. A parallel plate capacitor with large plates is charged and then disconnected from the battery. As the plates are pulled apart, does the potential difference increase, decrease or stay the same? How is the stored energy affected?

5. Give two reasons for using dielectrics in capacitors?

6. Water has a high dielectric constant. Why is it not frequently used in capacitors?

7. What is the difference between the dielectric constant and the dielectric strength?

8. Show that $\frac{F}{m} = \frac{C^2}{N.m^2}$

Practical activity III.3

Problems

1. In a parallel-plate capacitor the plates are separated by $0.8mm$. The plates carry charges $\pm 60nC$ and there is electric field of magnitude $3X10^4 \frac{V}{m}$ between the plates. Find (a) the potential difference (b) the capacitance (c) the plate area
2. The three capacitors in Fig.III.9a have an equivalent capacitance of $12.4\mu F$. Find C_1. The three capacitors in Fig.III.4(b) have an equivalent capacitance of $2.27\mu F$. Find C_3

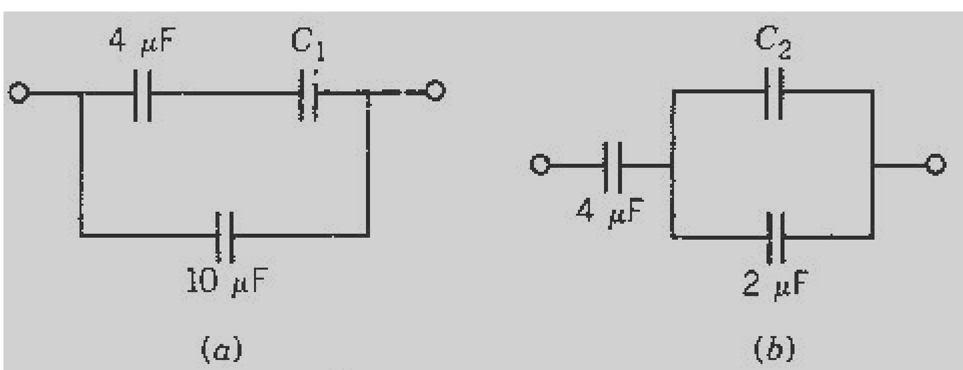


Fig.III.9 Problem III.2

3. A $50\mu F$ capacitor with an initial potential difference of $240V$ is completely discharged completely in $0.2ms$. What is the average power delivered?
4. A dielectric is inserted and completely fills the space between the plates of a capacitor. Determine the dielectric constant in each of the following cases: (a) The capacitance increases by 50% (b) The potential difference decreases by 25% (c) the stored energy doubles

5. Assume that the Earth ($R = 6400km$) is surrounded by a conducting sphere $50km$ above the surface and that there is a constant electric field of $100\frac{N}{C}$ directed vertically downward

   (i) What is the surface charge density of the system

   (ii) What is the capacitance of the system?

   (iii) Compare the answer to part (ii) with the capacitance of Earth treated as an isolated sphere

6. The space between the plates of a parallel-plate capacitor is filled with two dielectrics of equal size as shown in Fig.III.10. What is the resulting capacitance in terms of $\varepsilon_1, \varepsilon_2$ and C_0, the capacitance without the dielectric?

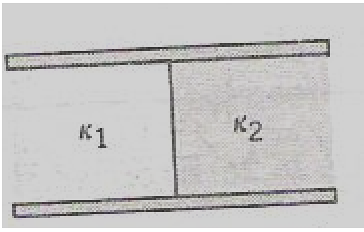


Fig.III.10. Problem 6

7. A cylindrical capacitor has a capacitance of $15pF$ for each $12cm$ of its length. The outer radius is $0.7cm$

   (i) What is the inner radius

   (ii) What is the linear charge density if $24V$ are applied across the plates?

8. A spherical drop of mercury has a capacitance of $0.05pF$. Two such drops collide and form a single larger spherical drop. What is the capacitance of the larger drop?

9. A parallel plate capacitor has a charge of 0.02μC on each plate with a potential difference of 240V. The plates are separated by 0.4mm of air.

   (i) What is the capacitance of the capacitor?

   (ii) What is the energy stored in the capacitor

   (iii) What is the area of a single plate?

   (iv) At what voltage will the air between the plates become ionized (electric breakdown)? Assume a dielectric strength of 3.0 kV/mm for air.

SI ELECTROMAGNETISM UNITS				
SYMBOL	**NAME OF QUANTITY**	**DERIVED UNITS**	**UNIT**	**BASE UNITS**
I	Electric current (intensity)	Ampere (SI Base Unit))	A	A
Q	Electric charge	Coulomb	C	A·s
U, ΔV, $\Delta\varphi$; E	Potential difference , Electromotive force	Volt	V	$kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$ (= J/C)
R; Z; X	(Electric) Resistance, Impedance, Reactance	Ohm	Ω	$kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$ (= V/A)
ρ	Resistivity	Ohm.metre	Ω·m	$kg \cdot m^3 \cdot s^{-3} \cdot A^{-2}$
P	(Electric) Power	Watt	W	$kg \cdot m^2 \cdot s^{-3}$ (= V·A)
C	Capacitance	Farad	F	$kg^{-1} \cdot m^{-2} \cdot s^4 \cdot A^2$ (= C/V)
E	Electric field (intensity, strength)	Volt per metre	V/m	$kg \cdot m \cdot s^{-3} \cdot A^{-1}$ (= N/C)
D	(Electric) Displacement (field)	Coulomb per square metre	C/m^2	$A \cdot s \cdot m^{-2}$
ε	Permittivity	Farad per metre	F/m	$kg^{-1} \cdot m^{-3} \cdot s^4 \cdot A^2$
χ_e	Electric susceptibility	(dimensionless)	–	–
G; Y; B	Conductance, Admittance, Susceptance	Siemens	S	$kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$ (= Ω^{-1})
κ, γ, σ	Conductivity	Siemens per metre	S/m	$kg^{-1} \cdot m^{-3} \cdot s^3 \cdot A^2$
B	Magnetic field vector, Magnetic flux density, magnetic induction	Tesla	T	$kg \cdot s^{-2} \cdot A^{-1}$ (= Wb/m^2 = $N \cdot A^{-1} \cdot m^{-1}$)
Φ	Magnetic flux	Weber	Wb	$kg \cdot m^2 \cdot s^{-2} \cdot A^{-1}$ (= V·s)
H	Magnetic field strength	Ampere per metre	A/m	$A \cdot m^{-1}$
L, M	Inductance	Henry	H	$kg \cdot m^2 \cdot s^{-2} \cdot A^{-2}$ (= Wb/A = V·s/A)
μ	Permeability	Henry per metre	H/m	$kg \cdot m \cdot s^{-2} \cdot A^{-2}$
χ	Magnetic susceptibility	(dimensionless)	–	–